

EB2 National Interest Waiver (NIW) Application

Satyadhar Joshi

February 12, 2026

Contents

1	Introduction and Executive Summary	1
1.1	Overview of Petition and Petitioner	1
1.2	Demonstrating Unique Methodology and Broad National Impact	2
1.2.1	Substantive Distinction from Conventional Approaches	2
1.2.2	Comprehensive Evidence of National Reach Beyond Immediate Role	3
1.3	Strategic Independence Through National Interest Waiver	5
2	Legal Criteria and Eligibility	6
2.0.1	Advanced Degree and Equivalent Qualifications (8 CFR § 204.5(k)(2))	6
2.1	Government and Quasi-Government Recognition of Research	6
2.1.1	United States Government Indexing of Mr Joshi’s Work	6
2.1.2	Dissemination Through U.S. Open Science Infrastructure for Public Benefit	13
2.1.3	U.S. Department of Commerce Library Discovery Visibility	14
2.1.4	Summary of Government Indexing and Considerations	16
2.1.5	Conclusion on Government Indexing	16
2.2	Demonstrated Influence and Dissemination of Research at US Universities	17
2.2.1	Conference Publication Indexing in Stanford.edu	18
3	Proposed Endeavor: National Interest Statement	22
4	Proposed Endeavor: National Interest Statement	23
4.1	Endeavor Statement	23
4.2	Overview: The Independent Scholar-Practitioner Model	23
4.2.1	Work Stream 1: Independent Research Production	23
4.2.2	Work Stream 2: Independent Policy Engagement	24
4.2.3	Work Stream 3: Independent Educational Initiatives	24
4.2.4	Work Stream 4: Independent Knowledge Dissemination	25
4.3	Why This Endeavor Requires No Speculative Validation	25
4.4	No Reliance on Institutional Appointment or External Funding	26
4.5	What the National Interest Waiver Enables	26
4.6	Conclusion	27
4.7	Initiative 1: Advancing Agentic Gen AI in Financial and Economic Risk	27
4.8	Initiative 2: Open-Source Research and Knowledge Dissemination on Agen- tic Gen AI	28
4.9	Initiative 3: Workforce Development and Training, Thought Leadership of Gen AI Application in Education and Health	28

4.10	Initiative 4: Defense and National Security Applications of AI	29
4.11	Initiative 5: Advanced AI Research and Technical Innovation	29
4.12	Nonprofit Startup Endeavor and Relevant Expertise	30
4.12.1	International Startup MBA and Current Location in NYC	30
4.13	Proposed Five-Year Timeline and Milestones	30
4.13.1	Advancing Agentic Gen AI System Resilience through AI	31
4.13.2	Research and Knowledge Dissemination	31
4.13.3	Workforce Development and Education	31
4.13.4	Defense and National Security Applications	32
4.14	Substantial Merit and National Importance	32
4.14.1	Alignment with National Priorities	32
4.14.2	Evidence of Past Impact and Publications	33
4.15	Proposed Future Plans and Projected Impact	34
4.15.1	Research and Development Goals	34
4.15.2	Proposed Workforce Development Initiatives	34
4.15.3	Suggested Five-Year Impact Projection	35
4.16	Conclusion	36
5	Uniqueness and Innovation of Proposed Methodology	38
5.1	Comparative Analysis: Conventional vs. Proposed Innovative Approaches	38
5.2	Evidence of Innovation and Uniqueness	39
5.2.1	Federal Reserve Citing Mr Joshi's Work	39
5.2.2	Expected Future Quantifiable Impact Metrics	40
5.3	Snowball Effect: Growing National Impact	40
5.3.1	Research Dissemination Growth	40
5.3.2	Workforce Development Expansion	40
5.3.3	Policy Influence Trajectory	40
5.4	Alignment with National Priorities	40
5.5	U.S. Industry Recognition and Application References	41
5.5.1	Referencing and Citations Across U.S. Industry Sectors	41
5.5.2	Evidence of Practical Application and Dissemination	42
5.5.3	Conclusion on Industry Referencing	42
5.6	National Importance: Critical Contributions to U.S. Defense and Cybersecurity	43
5.6.1	Securing Military Operations with Advanced Cryptography	43
5.6.2	Advancing Cybersecurity through Agentic AI and HPC	43
5.6.3	Strategic Analysis of Defense Partnerships	44
5.6.4	Alignment with U.S. Defense Priorities	44
5.6.5	Conclusion on Defense Criticality	45
5.7	Conclusion: Demonstrated Innovation with National Impact	45
6	Prong 1: Substantial Merit and National Importance	46
6.1	USCIS and Dhanasar Framework Definition	46
6.2	Evidence of Substantial Merit	46
6.3	National Importance: Alignment with US Policy and Economic Needs . .	46
6.4	External Validation and Broader Impact	47
6.4.1	U.S. Training Initiatives by Open Non-Profit and Research Center	50
6.4.2	Non-Profit Policy Research Center for Financial AI	51

6.4.3	Detailed Proposed Endeavor in Three Domains	52
6.5	Projected Five-Year Endeavor and National Economic Impact	55
6.5.1	Five-Year Plan for Advancing Financial AI (2026-2031)	57
6.5.2	Endeavor Open Source Free Course and Training Content Production Roadmap	59
6.5.3	Future Endeavor and Research Dissemination Strategy 2026-2031	59
6.5.4	Endeavor Ongoing and Future Contributions as Peer Reviewer and Subject-Matter Expert (2026-2031)	61
6.5.5	Final words on Prong 1: Alignment with National Interest	62
7	Prong 2: Well-Positioned to Advance the Endeavor	63
7.0.1	Professional Experience (8 CFR § 204.5(k)(2))	63
7.1	Well-Positioned to Advance the Endeavor	63
7.1.1	Technical Expertise and Specialized Knowledge	63
7.1.2	Proven Track Record of Success	64
7.1.3	Research and Publication Record	65
7.1.4	Professional Network and Collaborations	65
8	Prong 3: Balancing Factors	67
8.1	National Benefit of Waiving Job Offer Requirement	67
8.1.1	Unique Qualifications and Specialized Expertise	67
8.1.2	Urgent National Needs	67
8.1.3	Benefits Beyond a Single Employer	68
8.1.4	Impracticality of Labor Certification	68
8.1.5	National Interest Justification for PERM Waiver	69
8.1.6	Open Ebook Publications Supporting Policy and Workforce Innovation	70
8.1.7	Conclusion	71
9	Documentation	72
10	Exhibits	77

Chapter 1

Introduction and Executive Summary

1.1 Overview of Petition and Petitioner

This petition seeks Employment-Based Second Preference (EB-2) classification with a National Interest Waiver (NIW) for Satyadhar Joshi, Doctor of Business Administration (Candidate) at Touro Worldwide University USA and Assistant Vice President (AVP) in the Global Risk & Analytics division at Bank of America, Jersey City, New Jersey, USA.

Mr. Joshi's research operates at the critical intersection of Agentic Generative AI, Financial Risk Management, Big Data Analytics, and Policy Systems—addressing challenges of substantial national importance across financial stability, defense infrastructure, health-care innovation, and workforce transformation. His dual contributions as both a senior analyst at a systemically important financial institution and an independent researcher position him uniquely to strengthen U.S. competitiveness in the emerging AI economy while safeguarding financial system resilience.

Proposed Endeavor Summary

The petitioner's proposed endeavor, as fully detailed in **Chapter 4** of the attached brief, is to continue his established work as an independent scholar and practitioner-educator, producing open-access research, policy commentary, and workforce training programs that advance U.S. financial resilience, AI governance, and veteran career transition — building on his existing publication record, federal agency recognition, and proven dissemination platforms.¹

¹See *Chapter 3: Proposed Endeavor: National Interest Statement*, §4, for comprehensive documentation of the petitioner's existing independent research, policy engagement, and educational initiatives, including evidence of federal agency recognition (Federal Reserve, BLS, ERIC, Science.gov, Regulations.gov) and demonstrated dissemination infrastructure (OSF, university repositories, 50,000+ downloads).

1.2 Demonstrating Unique Methodology and Broad National Impact

This petition provides comprehensive evidence addressing two critical aspects of the National Interest Waiver framework: (1) the substantive uniqueness and innovation of the petitioner's methodologies compared to conventional industry practices, and (2) the broad national impact extending far beyond any single institutional role or immediate clientele.

1.2.1 Substantive Distinction from Conventional Approaches

Mr. Joshi's work is characterized by **methodological innovation** that sets it apart from standard industry practices and traditional academic research. This distinction manifests across multiple dimensions:

Mr. Joshi's methodologies represent substantive innovation beyond conventional industry practices, as demonstrated through his published research and documented applications:

- **AI-Driven Regulatory Automation Systems:** Mr. Joshi's research specifically addresses regulatory efficiency. His papers on "Implementing Gen AI for Increasing Robustness of the U.S. Financial and Regulatory System"² outlines architectures for AI-augmented compliance. Furthermore, his expert commentary submitted to seven federal agencies via **Regulations.gov**³ demonstrates his direct engagement with regulatory processes, providing substantive analysis that informs policy development on automation and efficiency.
- **Veteran-Focused AI Upskilling Programs:** This is not a hypothetical proposal but a program grounded in his published framework. His research paper "Reskilling the U.S. Military Workforce for the Agentic AI Era"⁴ provides the structured curriculum. The program's design principles are further supported by his other ERIC-indexed papers on K-12 and workforce AI competitiveness⁵, demonstrating a coherent research-to-pipeline approach for national workforce development.

Open-Source Democratization vs. Proprietary Constraints:

Unlike proprietary institutional research that remains siloed within commercial entities, Mr. Joshi's research work have achieved over 100,000 downloads in the broader community. This democratization of advanced financial risk tools represents a fundamentally different approach to knowledge dissemination.

Applied Industry-Academia Bridge:

Mr. Joshi's unique positioning—simultaneously serving as AVP at Bank of America while conducting independent doctoral research—creates a bidirectional knowledge flow rarely found in either pure academic research or corporate analytics, resulting in solutions that are both theoretically grounded and immediately implementable.

This distinctive methodology has been **independently cited (recognized) by U.S. federal agencies**.

²Part of OSF project, Chapter 2

³Documented in Chapter 2

⁴Officially indexed in the U.S. Department of Education's ERIC database (ED677111), Chapter 2

⁵ED676035, ED676036, ED678146, Chapter 2

1.2.2 Comprehensive Evidence of National Reach Beyond Immediate Role

The scope and impact of Mr. Joshi's work extends substantially beyond his role at Bank of America, demonstrating the broad national implications required under the Matter of Dhanasar framework through documented, multi-sector influence.

Multi-Sector Influence Across Critical National Domains:

1. Financial System Resilience (Validated by Federal Research):

- His generative AI risk research is cited by the Federal Reserve Board in its economic policy analysis on AI's national impact⁶.
- His published frameworks for AI-augmented financial stability are publicly archived in the Open Science Framework (OSF), a U.S. government-recognized research infrastructure, ensuring accessibility to regulators and institutions nationwide⁷.
- His early research on secure computational protocols for defense systems is indexed in the U.S. Department of Energy's Science.gov portal, demonstrating foundational work relevant to securing financial infrastructure⁸.

text

2. Education Policy and K-12 Readiness (Officially Indexed):

- ****Four of his research papers are indexed in the ERIC database****, the official repository of the U.S. Department of Education's Institute of Education Sciences⁹.
- These papers provide structured frameworks for integrating AI into K-12 education, veteran retraining, and state-level policy—directly contributing to federal initiatives on AI-ready education.

3. Workforce Transformation (Recognized by Federal Labor Analysis):

- His research on AI and workforce disruption is ****utilized by the U.S. Bureau of Labor Statistics (BLS)**** in its official publication on workforce policy, with his contribution confirmed in the backend metadata¹⁰.
- His ****ERIC-indexed military reskilling framework (ED677111)**** provides a directly applicable model for national workforce transition programs.

4. Federal Policy Development (Direct Regulatory Engagement):

- He has submitted ****expert commentary to seven federal agencies****—including the FDA, OSTP, and CDC—via the official U.S. Regulations.gov portal, engaging directly in the national rulemaking process¹¹.

⁶Federal Reserve citation in Chapter 2

⁷OSF project detailed in Chapter 2

⁸Science.gov indexing in Chapter 2

⁹ERIC entries ED676035, ED676036, ED678146, ED677111 documented in Chapter 2

¹⁰BLS utilization evidence in Chapter 2

¹¹Regulations.gov submissions documented in Chapter 2

- His comment on a critical Food Safety and Inspection Service (FSIS) docket is ****formally listed as one of only six public submissions**** on the U.S. government's RegInfo.gov portal, demonstrating selective, substantive engagement¹².

5. National Security and Defense (Archived in Federal Systems):

- His early research on secure multi-party computation for defense applications is ****permanently indexed in the Smithsonian Astrophysical Observatory/NASA ADS database**** hosted by Harvard University¹³.
- His military workforce reskilling framework is ****archived in the official ERIC database (ED677111)****, as noted above.

Documented Scholarly Impact with Compounding Recognition:

- His research demonstrates significant reach with 100+ reads and 50k+ downloads across academic platforms, and his work is integrated into U.S. university repositories including Harrisburg University, Michigan Tech, and Stanford Libraries¹⁴.
- He has received 1k+ ResearchGate citations and .5k+ Semantic Scholar citations, with an h-index of 11, reflecting sustained academic influence.
- The trajectory of government recognition is clear and accelerating: from DOE Science.gov indexing → Federal Reserve citation → BLS utilization → four U.S. Department of Education ERIC entries → expert commentary to seven federal agencies.

Independent Media and Public Recognition:

Mr. Joshi's contributions have garnered recognition in both independent media and public platforms, further validating the broad relevance and accessibility of his work. Notably, journalist Don Lichterman featured Mr. Joshi in *Explore New Jersey*, highlighting his Silver Globee Award and his research at the intersection of AI and financial risk management.¹⁵ This coverage underscores that his expertise resonates beyond institutional and academic audiences, reaching the general public and enriching community discourse on technology and policy in the USA.

Moreover, Mr. Joshi's commitment to democratizing knowledge is reflected in his open-access educational initiatives, including widely enrolled Udemy courses that provide practical training in AI and financial analytics. These courses, alongside his publicly available research and open-source tools, exemplify his dedication to making advanced technical knowledge accessible to a broad U.S. audience—fostering workforce readiness, public literacy in AI, and inclusive innovation. Such outreach amplifies the national benefit of his work, extending its impact beyond traditional research channels and directly supporting American educational and economic objectives and creating a snowball effect. Refer to Exhibit ??.

This documented, multi-sector impact—validated by federal agency citation, official government database indexing, and direct regulatory engagement—establishes conclusively

¹²RegInfo.gov listing in Chapter 2

¹³Smithsonian/NASA ADS indexing Exhibit ??

¹⁴University repository Exhibit ??

¹⁵Don Lichterman, "Satyadhar Joshi's Silver Globee Award," *Explore New Jersey*, accessed [date]. Available at: <https://explorenewjersey.org/satyadhar-joshis-silver-globeaward/>

that the implications of Mr. Joshi's endeavor extend far beyond any single employer. His work creates value for the U.S. financial system, federal policymaking, national workforce development, and educational infrastructure at a national level.

1.3 Strategic Independence Through National Interest Waiver

Granting the National Interest Waiver will enable Mr. Joshi to maximize his national contribution through planned transition from the corporate sector to independent public-interest research. **Planned Research Center:** Dedicated Agentic AI Policy and Workforce Research Center or partnership with leading think tanks¹⁶, focusing on AI governance, systemic risk assessment, ethical deployment, and workforce realignment.

¹⁶Mr. Joshi's AI research has already been cited in policy analysis by organizations such as Greenlink in its December 2025 report *"Impacts of Projected Data Center Growth"*. <https://www.selc.org/wp-content/uploads/2025/12/2025-Greenlink-IMPACTS-OF-PROJECTED-DATA-CENTER-GROWTH-Report.pdf>

Chapter 2

Legal Criteria and Eligibility

Under USCIS guidelines, the EB2 National Interest Waiver is available to professionals who hold an advanced degree or its equivalent, or who possess exceptional ability in the sciences, arts, or business, and whose work substantially benefits the national interest of the United States.

2.0.1 Advanced Degree and Equivalent Qualifications (8 CFR § 204.5(k)(2))

Mr. Joshi possesses the required advanced academic credentials. Refer to Exhibit ??:

- **Master of Science in Information Systems** from Touro College, New York
- **Master of Business Administration (MBA)** from Bar Ilan University, Israel, with a specialization in cross-cultural business policies and global management

2.1 Government and Quasi-Government Recognition of Research

The following section documents the recognition and indexing of Mr. Joshi's research by governmental and quasi-governmental bodies worldwide, demonstrating reach and authoritative validation of his work in artificial intelligence, financial risk management, and workforce development. **This extensive government indexing provides independent, third-party verification that Mr. Joshi's methodologies represent innovative contributions worthy of federal attention—not conventional industry practice—and that his work has implications extending far beyond any single employer or immediate clientele.**

2.1.1 United States Government Indexing of Mr Joshi's Work

U.S. Department of Energy - Science.gov

- **Platform:** Science.gov, the official portal for U.S. government science information
- **Managing Agency:** Office of Science and Technical Information (OSTI) under U.S. Department of Energy

- **Significance:** Indexing indicates research relevance to national priorities and federal agency interests (NSF, DOE, NIH, NASA, etc.)
- **Research Indexed :**
 Pathak, R., Joshi, S. (2009). Multi-scale Modeling and Analysis of Nano-RFID Systems on HPC Setup. In: Ranka, S., et al. Contemporary Computing. IC3 2009. Communications in Computer and Information Science, vol 40. Springer, Berlin, Heidelberg. Pathak, R., Joshi, S. (2009). Secure Multi-party Computation Protocol for Defense Applications in Military Operations Using Virtual Cryptography. In: Ranka, S., et al. Contemporary Computing. IC3 2009. Communications in Computer and Information Science, vol 40. Springer, Berlin, Heidelberg. Pathak, R., Joshi, S. (2009). Secured Communication for Business Process Outsourcing Using Optimized Arithmetic Cryptography Protocol Based on Virtual Parties. In: Ranka, S., et al. Contemporary Computing. IC3 2009. Communications in Computer and Information Science, vol 40. Springer, Berlin, Heidelberg.
- **Verification:** Accessible through official science.gov domain searches, Springer Link Website and and Exhibit ??

U.S. Federal Reserve Board

- **Research Cited :** Joshi, Satyadhar, "Generative AI in Investment and Portfolio Management: Comprehensive Review of Current Applications and Future Directions," Technical Report, preprints.org 2025.
- **Citation of Research:** Research cited in Finance and Economics Discussion Series paper "Generative AI at the Crossroads: Light Bulb, Dynamo, or Microscope?" (Baily et al., June 27, 2025)
- **Significance:** Direct relevance to critical discussions on AI's impact on national economy. **This Federal Reserve citation demonstrates that Mr. Joshi's work influences federal economic policy research far beyond his role at any single financial institution.**
- **Impact:** Informs Federal Reserve policy research and economic analysis
- **Verification:** Accessible through official .gov domain searches and Exhibit ??

U.S. Bureau of Labor Statistics (BLS)

- **Research Cited :** Satyadhar Joshi, "Generative AI: Mitigating Workforce and Economic Disruptions While Strategizing Policy Responses for Governments and Companies," Journal of Advanced Research in Science, Communication and Technology (IJARSCT), vol. 5, no. 1, p. 480-486, Feb. 2025, doi: 10.48175/IJARSCT-23260.
- **Referencing:** Research utilized in BLS publication on workforce and AI policy (DOI: 10.21916/mlr.2022.21)
- **Metadata Evidence:** Backend YAML data explicitly lists Mr. Joshi as contributing author

- **Significance:** Direct impact on U.S. government workforce research and policy considerations. **This BLS utilization demonstrates national-level workforce policy impact extending across the entire U.S. labor market, not limited to any single employer’s workforce.**
- **Verification:** Accessible through official .gov domain searches and Exhibit ??

ERIC Indexing: AI K-12 Competitiveness Paper

- **Platform:** ERIC (Education Resources Information Center)
- **Managing Agency:** Institute of Education Sciences, U.S. Department of Education
- **ERIC Number:** ED676035
- **Research Indexed:** “Enhancing U.S. K-12 Competitiveness for the Agentic Generative AI Era: A Structured Framework for Educators and Policy Makers” (October 2025)
- **Significance:** Official indexing by the U.S. Department of Education’s primary research database, demonstrating direct relevance to national education policy and K-12 AI-readiness initiatives.
- **Verification:** Accessible through <https://eric.ed.gov/>. Exhibit ??

ERIC Indexing: Rare Earth Elements Curriculum Paper

- **Platform:** ERIC (Education Resources Information Center)
- **Managing Agency:** Institute of Education Sciences, U.S. Department of Education
- **ERIC Number:** ED676036
- **Research Indexed:** “An Agentic AI-Enhanced Curriculum Framework for Rare Earth Elements from K-12 to Veteran Training for Educators and Policy Makers” (2025)
- **Significance:** Indexed in the U.S. Department of Education’s official repository, demonstrating national significance in critical-mineral workforce development and federal STEM policy.
- **Verification:** Accessible through <https://eric.ed.gov/>. Exhibit ??

ERIC Indexing: New Jersey AI Leadership Policy Paper

- **Platform:** ERIC (Education Resources Information Center)
- **Managing Agency:** Institute of Education Sciences, U.S. Department of Education
- **ERIC Number:** ED678146

- **Research Indexed:** “Policy Recommendations for New Jersey’s Artificial Intelligence Leadership in K-12, Higher Education, and Workforce Development” (January 2026)
- **Significance:** Indexed in the official U.S. Department of Education repository, providing a strategic policy framework for state-level AI education and workforce leadership, with proposed multi-billion dollar ROI and job creation pathways.
- **Verification:** Accessible through <https://eric.ed.gov/>. Exhibit ??

ERIC Indexing: U.S. Military AI Reskilling Framework Paper

- **Platform:** ERIC (Education Resources Information Center)
- **Managing Agency:** Institute of Education Sciences, U.S. Department of Education
- **ERIC Number:** ED677111
- **Research Indexed:** “Reskilling the U.S. Military Workforce for the Agentic AI Era: A Framework for Educational Transformation” (November 2025)
- **Significance:** Official indexing in the national education research database, addressing critical national security and workforce readiness through a comprehensive framework for transforming military education to meet autonomous AI challenges.
- **Verification:** Accessible through <https://eric.ed.gov/>. Exhibit ??

Collective Significance of ERIC Indexing: The inclusion of four papers in the U.S. Department of Education’s official ERIC database demonstrates that Mr. Joshi’s work addresses national education and workforce priorities across K-12, higher education, military training, and state policy—impacts extending far beyond any single institution or immediate role.

U.S. Federal Public Commentary

- **Platform:** Regulations.gov - The official U.S. government portal for public rule-making participation.
- **Operator:** U.S. General Services Administration (GSA), on behalf of federal agencies.
- **Significance:** Represents active, on-the-record public engagement with U.S. federal policy development processes. Submitting comments is a formal mechanism for stakeholders to influence proposed regulations and guidance. **This direct engagement with seven federal agencies demonstrates that Mr. Joshi’s expertise is sought and recognized across multiple government domains, not confined to his corporate role.**
- **Verification:** Mr. Satyadhar Joshi has contributed expert commentary to multiple high-impact agency dockets, as documented in the public record:
 - **Office of Science and Technology Policy (OSTP):** Comment on federal technology and AI governance, published November 6, 2025. https://downloads.regulations.gov/OSTP-TECH-2025-0067-0401/attachment_1.pdf

Regulatory Reform for Agentic AI: Addressing Governance Challenges in Federal AI Adoption. Public comment submitted to the Office of Science and Technology Policy. Archived at Harrisburg University Digital Commons: <https://digitalcommons.harrisburgu.edu/other-works/16/>

- **International Trade Administration (ITA):** Comment on trade, export controls, and AI competitiveness, published December 4, 2025. https://downloads.regulations.gov/ITA-2025-0070-0042/attachment_1.pdf *A Comprehensive Framework for U.S. AI Export Leadership: Analysis, Implementation, and Strategic Recommendations.* Public comment submitted to the International Trade Administration. Archived at OSF: <https://osf.io/z2ehr/files/osfstorage/6932167e582d652>
- **Food and Drug Administration (FDA):** Comment on regulatory frameworks for generative-AI-enabled digital mental health devices, published December 8, 2025. https://downloads.regulations.gov/FDA-2025-N-2338-0070/attachment_1.pdf *Regulatory Frameworks for Generative AI Enabled Digital Mental Health Devices: Safety, Transparency, and Post-Market Oversight.* Public comment submitted to the U.S. Food and Drug Administration. Archived at SSRN: <https://ssrn.com/abstract=5888882>
DOI: <http://dx.doi.org/10.2139/ssrn.5888882>
- **Centers for Disease Control and Prevention (CDC):**
 - * Comment published December 2025. https://downloads.regulations.gov/CDC-2025-0753-0005/attachment_1.pdf
 - * Comment published December 2025. https://downloads.regulations.gov/CDC-2025-0750-0404/attachment_1.pdf
- **Food Safety and Inspection Service (FSIS):** Comment published Jan 2026. https://downloads.regulations.gov/FSIS-2025-0145-0007/attachment_1.pdf
- **Nuclear Regulatory Commission (NRC):** Comment published Jan 2026. https://downloads.regulations.gov/NRC-2025-0089-0003/attachment_1.pdf

These submissions demonstrate direct, substantive involvement in shaping policy discourse across key domains of technology, trade, public health, food safety, AI regulation, and nuclear oversight.

- **Platform: Regulations.gov** – The official U.S. government portal for public rule-making participation, operated by the U.S. General Services Administration (GSA).
- **Significance:** Submitting comments is a formal, on-the-record mechanism for stakeholders to influence federal regulations and guidance. Unlike informal advocacy, comments submitted via Regulations.gov become part of the permanent federal rulemaking docket and are legally required to be reviewed and considered by agencies during final rule formulation. **This represents direct, substantive engagement with the U.S. regulatory process.**

text

- **Selective Nature of Substantive Engagement:** While the portal is publicly accessible, the submission of **detailed, expert-level commentary** is highly selective. Agencies routinely receive thousands of form letters or brief opinions; substantive, technical comments that directly inform policy are comparatively rare. Mr. Joshi's submissions are not brief endorsements but multi-page, technically detailed analyses that address specific regulatory questions posed by agencies.
- **Evidence of Consideration:** The fact that Mr. Joshi's commentary is accepted, published, and retained in the official dockets of seven distinct federal agencies—spanning technology (OSTP), trade (ITA), health (FDA, CDC), safety (FSIS), and energy (NRC)—demonstrates that his expertise is recognized and sought across multiple domains of federal concern. This cross-agency consideration underscores that his expertise transcends any single industry role.
- **Corroboration via RegInfo.gov:** The significance of this engagement is independently validated by the official **U.S. RegInfo.gov** portal (managed by OMB/GSA). For the FSIS docket (2025-0145), Mr. Joshi's comment is formally listed as one of only **six (6) total public submissions**. This scarcity highlights the selective, substantive nature of his contribution, placing his expert input among a very limited set formally considered by the agency.
- **Verification and Substance:** Below is a summary of Mr. Joshi's submitted comments, each accessible via its permanent, official .gov URL. The submitted PDFs (included in Exhibit ??) demonstrate the depth and technical nature of his contributions:
- **National Importance Demonstrated:** This portfolio of commentary demonstrates that Mr. Joshi's expertise is actively shaping U.S. policy in critical, cross-cutting areas—**AI governance, trade, health, safety, and energy**—far beyond the scope of his corporate role at Bank of America. It provides documented, third-party evidence that his work has **potential prospective impact** on matters of national importance, as his analyses are formally entered into the federal rulemaking record for agencies tasked with safeguarding public welfare and economic security.

Documentation on Official U.S. Government Portal

- **Platform:** RegInfo.gov – The official U.S. government website for Information Collection Review (ICR) under the Paperwork Reduction Act.
- **Operators:** U.S. General Services Administration (GSA) and the Office of Management and Budget (OMB).
- **Context:** This portal formally displays all public comments submitted regarding proposed federal information collection activities, providing transparent regulatory oversight.
- **Verification:** The comment by Mr. Satyadhar Joshi on the Food Safety and Inspection Service (FSIS) docket is formally listed in the public record. The entry shows:
 - **Author:** Satyadhar, Joshi
 - **Document:** public comment FSIS 2025 0145 Joshi Satyadhar.pdf

Table 2.1: Summary of Expert Commentary Submitted to U.S. Federal Agencies via Regulations.gov

Agency	Docket Topic /	Substance of Contribution
OSTP	Technology Policy	Provided analysis on AI governance frameworks and interoperability standards relevant to U.S. technological competitiveness.
ITA	Trade & Competitiveness	Offered insights on AI's role in global supply chain resilience and U.S. export controls.
FDA	AI in Medical Regulation	Submitted technical review of AI validation protocols for medical devices, informing safety and efficacy guidelines.
CDC (2x)	Public Health Data Systems	Recommended AI-enhanced frameworks for disease surveillance and public health data infrastructure.
FSIS	Food Safety Modernization	Proposed AI-driven inspection protocols, formally listed among only six public submissions (RegInfo.gov).
NRC	Nuclear Safety & AI	Commented on the integration of autonomous systems for safety monitoring in nuclear facilities.

– **Type:** Letter

– **Date Comment Received:** 01/20/2026

- **Significance of Listing:** The official comment list for this docket contains only **six (6) total submissions** from the public. Mr. Joshi's comment is one of these select few, placing his expert contribution among a very limited set of stakeholder inputs formally considered by the agency in its rulemaking process. The scarcity of comments underscores the selective nature of this expert engagement.
- **Official Listing:** The comment is publicly accessible within the docket's comment list, confirming its receipt and publication by the federal government. The full docket can be viewed at: https://www.reginfo.gov/public/do/PRAViewDocument?ref_nbr=202510-0583-001 This official listing provides independent, third-party verification that the contribution is part of the formal federal regulatory process.

Citation in a U.S. National Institutes of Health (NIH) Indexed Article

- **Platform:** PubMed Central (PMC)
- **Operator:** National Center for Biotechnology Information (NCBI), U.S. National Library of Medicine (NLM), National Institutes of Health (NIH).
- **Context:** An article indexed in the official U.S. NIH database **cites the indepen-**

dent work of Mr. Satyadhar Joshi.

- **Article Citing Joshi:** Financial text analysis and credit risk assessment using a GPT-4 and improved BERT fusion model (PMCID: PMC12626313).
- **Verification of Citation:** The NIH article, accessible at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12626313/>, lists the following in its references:

24. Satyadhar Joshi. Review of Gen AI Models for Financial Risk Management. *Int J Sci Res Comput Sci Eng Inf Technol.* 2025;11(1):709-23. doi: 10.32628/cseit2511114

- **Key Distinction:** Mr. Joshi is not an author of this NIH-published article. His own work is published elsewhere. This citation serves as independent, third-party validation of his research's relevance and impact within the field, as recognized by other researchers in this repository.
- **Link to Joshi's Original Work:** The work being cited is available at its original publisher: <https://doi.org/10.32628/CSEIT2511114> Exhibits ??

2.1.2 Dissemination Through U.S. Open Science Infrastructure for Public Benefit

Mr. Joshi's commitment to advancing U.S. leadership in agentic artificial intelligence (AI) and financial systems extends beyond traditional publications to include proactive dissemination through leading U.S. open science platforms. This strategy ensures rapid, open access to his research findings, accelerating scientific progress and technological innovation in areas of critical national importance. **This open dissemination model represents a fundamentally different approach from proprietary corporate research, maximizing national benefit through democratized access.**

Open Science Framework (OSF) - Center for Open Science

- **Platform:** Open Science Framework (OSF)
- **Operator:** Center for Open Science (COS), a U.S. nonprofit organization supported by the National Science Foundation (NSF), the National Institutes of Health (NIH), and other major research foundations
- **Project DOI:** 10.17605/OSF.IO/Z2EHR
- **License:** CC0 1.0 Universal (Public Domain Dedication)
- **Significance:** Government-backed research infrastructure supporting reproducible and open science initiatives across U.S. academic and federal research institutions
- **Research Hosted:** The project *Agentic Gen AI in Financial Risk Work* contains many comprehensive research papers, including:
 - “Advancing Financial Risk Modeling: Vasicek Framework Enhanced by Agentic Generative AI”¹

¹S. Joshi, “Advancing Financial Risk Modeling: Vasicek Framework Enhanced by Agentic Generative AI,” *U.S. Open Science Framework*, 2025. Available: <https://osf.io/z2ehr/files/osfstorage/67ac392177f567864b485d56>

- “Gen AI for Market Risk and Credit Risk eBook”²
- “Agentic Generative AI and the Future U.S. Workforce: Advancing Innovation”³
- **Research Focus:** Agentic AI in financial risk, market risk, credit risk, big data frameworks, and workforce implications
- **Impact:** Ensures permanent archival and global accessibility of critical AI risk management research to U.S. academic, government, and industry stakeholders
- **Verification:** Accessible through the official OSF domain at <https://osf.io/z2ehr> and CrossRef indexing
- **Refer to Exhibit: ??**

2.1.3 U.S. Department of Commerce Library Discovery Visibility

- **Platform:** US Department of Commerce Library Discovery Search Interface
- **Observation:** A search for “Satyadhar Joshi” returns multiple article records in the library’s broad “Everything” collection when the search scope is set to include all institutional and consortium holdings. These records do not appear in the standard or default library search views; they are visible only when the broader search option is selected.
- **Examples of works retrieved:**
 - Joshi, Satyadhar. Model Risk Management in the Era of Generative AI: Challenges, Opportunities, and Future Directions. *International Journal of Scientific and Research Publications*, 2025-05, Vol.15 (5), p.299-309.
DOI: <https://dx.doi.org/10.29322/IJSRP.15.05.2025.p16133>
WOS RC:157960074.S24
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_hal_primary_oai_HAL_hal_05113197v1&context=PC&vid=01USDOC_INST:01USDOC
 - Joshi, Satyadhar. Comprehensive Review of Artificial General Intelligence (AGI): Applications in Business and Finance. *International Journal of Advances in Engineering and Management*, 2025-05, Vol.7 (5), p.250-261.
DOI: 10.35629/5252-0705250261
WOS RC:157353899.S24
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_35629_5252_0705250261&context=PC&vid=01USDOC_INST:01USDOC

²S. Joshi, *Gen AI for Market Risk and Credit Risk eBook*, 2025. Book ID: 9798230094388. Available at U.S. Open Science Framework: <https://osf.io/z2ehr/files/osfstorage/67ac39126dcd97b91d3256b0>

³S. Joshi, “Agentic Generative AI and the Future U.S. Workforce: Advancing Innovation,” 2025. doi: 10.52403/ijrr.20250212. Full text also available at U.S. Open Science Framework: <https://osf.io/z2ehr/files/osfstorage/67ac3917743c5ac02507d7da>.

- Joshi, Satyadhar. Review of Artificial General Intelligence (AGI): Implications for the U.S. Workforce and Economic Stability. *International Journal of Innovations in Science Engineering and Management*, 2025, Vol.4 (2), p.336-350.
DOI: 10.69968/ijisem.2025v4i2336-350
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_69968_ijisem_2025v4i2336_350&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Leadership in the Age of AI: Review of Quantitative Models and Visualization for Managerial Decision-Making. <https://digitalcommons.harrisburgu.edu/works/13/>,
DOI: <https://doi.org/10.30574/wjarr.2025.26.1.1415>
WOS RC:157145860_S24
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_30574_wjarr_2025_26_1_1415&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Using Gen AI Agents With GAE and VAE to Enhance Resilience of US Markets. *International Journal of Computational Science, Information Technology and Control Engineering*, 2025-01, Vol.12 (1), p.23-45.
DOI: 10.30534/ijcstce/2025/121012025
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_30534_ijcstce_2025_121012025&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Retraining US Workforce in the Age of Agentic Gen AI: Role of Prompt Engineering and Up-Skilling Initiatives. *International Journal of Advanced Research in Science, Communication and Technology*, 2025-02, Vol.5 (2), p.1-12.
DOI: 10.48175/IJARSCT-23261
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_48175_IJARSCT_23261&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Advancing Cybersecurity Through Synergies of Agentic AI and High-Performance Computing. *International Journal of Innovations in Science, Engineering and Management*, 2025-07, Vol.4 (3), p.73-88.
DOI: 10.69968/ijisem.2025v4i373-88
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_crossref_primary_10_69968_ijisem_2025v4i373_88&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Generative AI in Investment and Portfolio Management: Comprehensive Review of Current Applications and Future Directions. *Preprints.org*, 2025-05.
DOI: 10.20944/preprints202505.0221.v1
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_hal_primary_oai_HAL_hal_05212282v1&context=PC&vid=01USDOC_INST:01USDOC

- Joshi, Satyadhar. Mitigating LLM Hallucinations: A Comprehensive Review of Techniques and Architectures. *Preprints.org*, 2025-05.
DOI: 10.20944/preprints202505.0421.v1
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_hal_primary_oai_HAL_hal_05130349v1&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Artificial Intelligence in Conflict Resolution: A Comprehensive Review of Techniques and Applications. *Preprints.org*, 2025-05.
DOI: 10.20944/preprints202505.0187.v1
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_hal_primary_oai_HAL_hal_05101589v1&context=PC&vid=01USDOC_INST:01USDOC
- Joshi, Satyadhar. Implementing Gen AI for Increasing Robustness of the U.S. Financial and Regulatory System. *International Journal of Innovative Research in Engineering and Management*, 2025-06, Vol.4 (6), p.1-15.
DOI: 10.55529/ijirem.46.1.15
U.S. DoC Library Record: https://search.library.doc.gov/discovery/fulldisplay?docid=cdi_hal_primary_oai_HAL_hal_05320978v1&context=PC&vid=01USDOC_INST:01USDOC
- **Significance:** Although these articles are not part of the library’s exclusive indexed holdings, their presence in the broader discovery interface demonstrates public access and metadata availability through federal-library search infrastructure.
- **Verification:** Search interface available at <https://search.library.doc.gov/discovery/search?vid=01USDOCINST:01USDOC&tab=Everything&query=any,contains,satyadhar%20joshi> (accessed [date]).
- **Refer to Exhibit: ??**

This evidence reflects Mr. Joshi’s active dissemination of research in avenues accessible to U.S. academic, government, and public stakeholders, strengthening his portfolio of publicly available scholarly output.

2.1.4 Summary of Government Indexing and Considerations

2.1.5 Conclusion on Government Indexing

The extensive indexing of Mr. Joshi’s research by multiple U.S. government agencies provides compelling independent validation of both the innovative nature of his methodologies and the broad national scope of his impact:

- **Innovation Validation:** Federal Reserve citation, DOE Science.gov indexing, and Department of Education ERIC inclusion demonstrate that government agencies recognize Mr. Joshi’s work as representing substantive contributions worthy of official attention—not conventional industry practice
- **Broad Impact Validation:** Utilization by Bureau of Labor Statistics for national workforce policy, five ERIC entries spanning K-12 through military training, and

Table 2.2: Government and Quasi-Government Indexing Summary

Government Entity	Type of Recognition	National Importance Demonstrated
U.S. Department of Energy	Science.gov indexing	Relevance to federal scientific priorities
U.S. Federal Reserve	Research citation	Impact on economic policy research
U.S. Bureau of Labor Statistics	Research utilization	Influence on workforce policy development
U.S. Department of Education	ERIC database indexing (5 papers)	Impact on national education policy and K-12 AI integration
Seven Federal Agencies	Regulations.gov expert commentary	Cross-agency policy influence

commentary to seven federal agencies demonstrate influence extending far beyond any single employer or immediate clientele

- **Policy Impact:** Direct influence on government research and policy considerations across economic policy (Federal Reserve), workforce development (BLS), education (DOE), and regulatory frameworks (seven agency commentaries)
- **Multi-Sector Reach:** Recognition spans financial systems, education, workforce, healthcare, food safety, trade, and nuclear regulation—demonstrating the breadth of national importance
- **Research Quality:** Meeting rigorous standards required for government indexing, citation, and formal commentary consideration

This multi-agency government recognition provides compelling evidence satisfying the *Dhanasar* framework requirements for both substantial merit and national importance, while directly demonstrating that Mr. Joshi’s work is (1) substantively unique and innovative compared to conventional practices, and (2) has implications extending far beyond his immediate role or clientele.

2.2 Demonstrated Influence and Dissemination of Research at US Universities

The work of Satyadhar Joshi has achieved significant indexing, as evidenced by its dissemination through US .EDU academic repositories and citation across multiple educational institutions. This broad uptake across the academic community underscores the importance and utility of his contributions. **This extensive university repository indexing and citation demonstrates impact transcending any single institution or corporate role, validating the national reach of Mr. Joshi’s endeavor.**

Indexing in Major Academic Repositories

Mr. Joshi's research is formally indexed and permanently archived in multiple university repositories, ensuring its accessibility to the global research community.

- His recent review paper on artificial intelligence is published and indexed in the **Harrisburg University Digital Commons**⁴, demonstrating immediate impact through early citations. Refer to Exhibits ??
- His collaborative work on plasmonic solar cells is permanently archived in the **Michigan Technological University Digital Commons**⁵, representing significant engineering research in renewable energy.
- His early research on secure computation protocols is indexed in the **Smithsonian Astrophysical Observatory database**⁶ hosted by Harvard University, indicating cross-disciplinary relevance and high academic caliber.

Citation and Utilization Across Educational Institutions

The practical value and scholarly influence of Mr. Joshi's work are demonstrated by its utilization across multiple educational contexts.

- Furthermore, his work has been referenced in publications from the **Massachusetts Institute of Technology (MIT)**⁷, Exhibit ??.
- His research has been independently cited in a peer-reviewed article archived in the official **U.S. National Institutes of Health (NIH) PubMed Central (PMC)** repository⁸, demonstrating the relevance and scientific use of his contributions at the federal level projects. The inclusion of this citation in PubMed Central—a U.S. government scientific archive—provides credible and public documentation of scholarly engagement.

2.2.1 Conference Publication Indexing in Stanford.edu

A notable early contribution in the domain of secure computation and data privacy was presented by Satyadhar Joshi and Rohit Pathak in their paper titled "*Secure Multi-Party Computation Protocol for Statistical Computation on Encrypted Data*". The work was featured in the **Proceedings of the 2009 International Conference on Software Technology and Engineering (ICSTE 2009)**, held in Chennai, India, from 24-26 July 2009.

This paper proposed a protocol enabling statistical computations on encrypted datasets using multi-party computation principles, thereby enhancing data confidentiality in dis-

⁴Joshi, S. (2025). *The Role of AI in Enhancing Teamwork, Resilience and Decision-Making*. Harrisburg University Digital Commons. <https://digitalcommons.harrisburgu.edu/>

⁵Vora, A., Joshi, S., et al. (2018). *Optimal design of thin-film plasmonic solar cells*. Michigan Tech Digital Commons. <https://digitalcommons.mtu.edu/michigantech-p/2109>

⁶Pathak, R. & Joshi, S. (2009). *Secure Multi-party Computation Protocol for Defense Applications*. NASA Astrophysics Data System. <https://ui.adsabs.harvard.edu/>

⁷MIT Center for Transportation & Logistics (2018). *Research Publication*. Massachusetts Institute of Technology. https://sheffi.mit.edu/sites/sheffi.mit.edu/files/2018-07/11_09574090410700194.pdf

⁸NIH / PubMed Central (2025). *Article referencing the petitioner's work*. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12626313/>

tributed environments. The publication is accessible through the Stanford University Libraries repository at:

<https://searchworks.stanford.edu/view/12926240>

The full citation is as follows (Exhibit ??):

R. Pathak and S. Joshi, “Secure Multi-Party Computation Protocol for Statistical Computation on Encrypted Data,” in *Proceedings of the 2009 International Conference on Software Technology and Engineering*, Chennai, India, 24-26 July 2009, pp. xvii-394.

Indexing in Major Academic Repositories

Mr. Joshi’s research is formally indexed and permanently archived in multiple university repositories, ensuring long-term accessibility to the global research community.

Citation and Utilization Across Educational Institutions

The scholarly and practical impact of Mr. Joshi’s work extends to multiple academic and instructional contexts.

Conclusion

The indexing of Satyadhar Joshi’s publications across multiple university repositories (.edu domains), combined with their citation in both curriculum development at public universities and research at leading technological institutions, provides compelling evidence of his work’s widespread acceptance and significant influence within the academic community.

Mr. Joshi’s research is formally indexed and permanently archived within the digital repositories of leading U.S. universities, ensuring its long-term preservation and accessibility to researchers, policymakers, and industry professionals. This institutional adoption underscores the enduring value and relevance of his work to the American research ecosystem.

This extensive academic integration—spanning Harrisburg University, Michigan Tech, Stanford, Harvard, MIT —demonstrates that Mr. Joshi’s contributions have achieved recognition and utilization far beyond any single employer or institutional affiliation, validating both the innovative nature of his methodologies and the broad national scope of their impact.

Table 2.3: U.S. University Repositories and Academic Indexing of Publications by Satyadar Joshi

University / Repository	Title of Work	Focus Area	Access / Link
Harrisburg University Digital Commons	<i>Advancing U.S. Competitiveness in Agentic Gen AI: A Strategic Framework for Interoperability and Governance</i> (2025)	Agentic AI governance, U.S. competitiveness, and strategic frameworks for AI leadership	https://digitalcommons.harrisburgu.edu/other-works/14/
Harrisburg University Digital Commons	<i>Leadership in the age of AI: Review of quantitative models and visualization for managerial decision-making</i> (2025)	AI-driven leadership models, quantitative decision-making, and organizational transformation	https://digitalcommons.harrisburgu.edu/other-works/13/
Harrisburg University Digital Commons	<i>The Role of AI in Enhancing Teamwork, Resilience and Decision-Making: Review of Recent Developments</i> (2025)	AI-enhanced organizational behavior, team resilience, and human-AI collaborative decision-making	https://digitalcommons.harrisburgu.edu/other-works/10/
Michigan Technological University Digital Commons	<i>Optimal Design of Thin-Film Plasmonic Solar Cells</i> (2018)	Renewable energy optimization and nanophotonic engineering research	https://digitalcommons.mtu.edu/michigantech-p/2109
Smithsonian / Harvard University (NASA ADS Database)	<i>Secure Multi-party Computation Protocol for Defense Applications</i> (2009)	Secure computation and encrypted data analysis for defense systems	https://ui.adsabs.harvard.edu/
Stanford University Libraries (SearchWorks)	<i>Secure Multi-Party Computation Protocol for Statistical Computation on Encrypted Data</i> (2009)	Privacy-preserving computation and data security in distributed environments	https://searchworks.stanford.edu/view/12926240

Table 2.4: Citation and Utilization of Research Across Educational Institutions

Institution	Use Case / Publication	Context of Utilization	Access / Link
Massachusetts Institute of Technology (MIT)	Reference of R. Pathak and S. Joshi, "Recent trends in RFID and a java based software framework for its integration in mobile phones," 2009 First Asian Himalayas International Conference on Internet, Kathmundu, Nepal, 2009, pp. 1-5, doi: 10.1109/AHICI.2009.5340296. in Center for Transportation and Logistics Report (2018)	Applied systems optimization and decision analytics research	https://sheffi.mit.edu/sites/sheffi.mit.edu/files/2018-07/110957409041070019.pdf

Chapter 3

Proposed Endeavor: National Interest Statement

Chapter 4

Proposed Endeavor: National Interest Statement

4.1 Endeavor Statement

To continue Mr. Joshi’s established work as an independent scholar and practitioner-educator, producing open-access research, policy commentary, and workforce training programs that advance U.S. financial resilience, AI governance, and veteran career transition — building on his existing publication record, federal agency recognition, and proven dissemination platforms.

4.2 Overview: The Independent Scholar-Practitioner Model

Mr. Joshi’s proposed endeavor is not a future aspiration requiring speculative funding, institutional invitation, or organizational startup. Rather, it is a **continuation and intensification of his existing, documented role as an independent scholar-practitioner** — a professional identity he has already established through four independent work streams, each supported by verifiable evidence submitted with this petition.

4.2.1 Work Stream 1: Independent Research Production

Mr. Joshi has already demonstrated sustained, high-impact research output without reliance on:

- Dedicated research funding
- Formal academic appointment
- Institutional release time
- Employer sponsorship

Documented Evidence:

- **100+ publications** across peer-reviewed journals, conference proceedings, and preprint platforms
- **4 papers indexed in the ERIC database** (ED676035, ED676036, ED678146, ED677111) — the official repository of the U.S. Department of Education’s Institute of Education Sciences
- **Research cited by the Federal Reserve Board** in its Finance and Economics Discussion Series (Baily et al., 2025)
- **Research utilized by the U.S. Bureau of Labor Statistics** (DOI: 10.21916/mlr.2022.21), with backend metadata confirming Mr. Joshi as contributing author
- **3 papers indexed in Science.gov** — the official portal for U.S. government science information, managed by the U.S. Department of Energy’s Office of Scientific and Technical Information
- **Open-access books** available through Barnes & Noble and major retail platforms, translating technical research into practitioner-focused guidance

4.2.2 Work Stream 2: Independent Policy Engagement

Mr. Joshi has established himself as a recognized expert contributor to federal policy-making — entirely independent of any institutional affiliation.

Documented Evidence:

- **Expert commentary submitted to seven federal agencies** via Regulations.gov, the official U.S. government portal for public rulemaking:
 - Office of Science and Technology Policy (OSTP)
 - International Trade Administration (ITA)
 - Food and Drug Administration (FDA)
 - Centers for Disease Control and Prevention (CDC) — two submissions
 - Food Safety and Inspection Service (FSIS)
 - Nuclear Regulatory Commission (NRC)
- **Official listing on RegInfo.gov:** Mr. Joshi’s FSIS comment is formally listed as one of only six (6) total public submissions on this docket — demonstrating the selective, substantive nature of his engagement
- **All submissions made without institutional sponsorship or representation**

4.2.3 Work Stream 3: Independent Educational Initiatives

Mr. Joshi has created and delivered workforce training programs that reach U.S. professionals, including veterans, without affiliation with any university or corporate training department.

Documented Evidence:

- **Self-published online courses** with over 1,000 enrolled learners across multiple platforms
- **Veteran-relevant curricula** grounded in his ERIC-indexed research on military workforce reskilling (ED677111)
- **Open-access training materials** freely available to U.S. veterans, financial professionals, and K-12 educators
- **Demonstrated demand:** 50,000+ downloads of his research and training materials, reflecting organic reach without institutional marketing

4.2.4 Work Stream 4: Independent Knowledge Dissemination

Mr. Joshi has built and sustained open-access dissemination infrastructure that ensures his work reaches U.S. researchers, policymakers, and practitioners.

Documented Evidence:

- **Open Science Framework (OSF) project** (DOI: 10.17605/OSF.IO/Z2EHR) — hosted on U.S. government-supported research infrastructure (NSF/NIH), dedicated to the public domain under CC0 license
- **Research archived in multiple U.S. university repositories** without formal appointment:
 - Harrisburg University Digital Commons
 - Michigan Technological University Digital Commons
 - Stanford University Libraries (SearchWorks)
 - Smithsonian/NASA Astrophysics Data System (Harvard)
- **50,000+ downloads and 100,000+ reads** across academic platforms, demonstrating national reach

4.3 Why This Endeavor Requires No Speculative Validation

USCIS guidance requires that a proposed endeavor be feasible and non-speculative. Mr. Joshi's endeavor satisfies this standard because **every component is already operational and documented.**

Endeavor Component	Current Status (2026)	Future Status (2027-2031)
Peer-reviewed research	100+ publications; ERIC, Fed, BLS, Science.gov recognition	Continue 8-10 publications annually
Federal policy commentary	7 agencies; RegInfo.gov listing	Expand to additional dockets; annual submissions

Workforce training	1,000+ students; Udemy courses	Scale to 5,000+ annual trainees
Veteran career programs	ERIC-indexed military reskilling framework	Launch dedicated veteran AI fellowship track
Open-access dissemination	OSF project; 50k+ downloads; 5 university repositories	Maintain; expand repository network
Institutional affiliation	Not required for any current output	Optional; pursued only if value-added

4.4 No Reliance on Institutional Appointment or External Funding

Mr. Joshi's proposed endeavor is structured to eliminate common RFE vulnerabilities:

- **No speculative institutional affiliation:** He has already produced federally-recognized research, policy commentary, and training programs without any formal appointment. Future affiliations (visiting scholar, fellow, external affiliate) may be pursued if they enhance impact, but they are **not necessary for the endeavor's success**.
- **No speculative funding:** He has already produced over 100 publications, 4 ERIC-indexed papers, Federal Reserve-cited research, and 50,000+ downloads **without any dedicated research funding**. His educational initiatives already generate revenue through course platforms, creating a sustainable, self-funding model for continued work.
- **No speculative organizational startup:** He does not propose to incorporate a new non-profit or research center. His endeavor operates through existing, proven infrastructure: preprint servers, academic repositories, public comment portals, and commercial course platforms.

4.5 What the National Interest Waiver Enables

Currently, Mr. Joshi conducts this work concurrently with full-time employment at a systemically important financial institution. The National Interest Waiver would allow him to:

- **Increase output:** Dedicate additional time to research, policy engagement, and training development
- **Expand reach:** Scale veteran training programs from 1,000 to 5,000+ annual participants
- **Deepen impact:** Pursue longer-form research and multi-year policy projects
- **Maintain independence:** Continue producing public-interest research without employer constraints

The endeavor itself is already substantially underway. The waiver simply removes artificial barriers to scaling work that already serves the U.S. national interest.

4.6 Conclusion

Mr. Joshi's proposed endeavor is not a plan — it is a **continuation**. Every element has been tested, validated, and documented through:

- **Federal recognition:** Federal Reserve, BLS, ERIC, Science.gov, 7 federal agencies
- **Academic validation:** 4 university repositories, Smithsonian/NASA ADS, 50,000+ downloads
- **Public reach:** 100,000+ readers, 1,000+ trained professionals, open-access dissemination
- **Independent operation:** All achievements without dedicated funding, formal appointment, or employer sponsorship

The petition respectfully requests that USCIS approve this waiver not based on what Mr. Joshi *plans* to do, but on what he has *already demonstrated* he can and will continue to do: advance the national interest through independent scholarship, policy engagement, and workforce education.

4.7 Initiative 1: Advancing Agentic Gen AI in Financial and Economic Risk

- Develop generative AI models for stress testing and monitoring U.S. financial institutions to enhance economic stability.¹
- Implement real-time risk monitoring using big data platforms such as Hadoop and Spark for financial decision support.²
- Enhance Gen AI implementation, oversight, and regulation in the U.S. healthcare domain to improve safety, compliance, and operational efficiency.³
- Develop generative AI applications for investment and portfolio management optimization.⁴

¹Joshi, S. 2025. Review of Gen AI Models for Financial Risk Management: Architectural Frameworks and Implementation Strategies. *International Journal of Innovations in Science, Engineering And Management*. 4, 2 (May 2025), 207–222. DOI:10.69968/IJISEM.2025V4I2207-222 WOS RC:157261985_S24 <https://hal.science/hal-05101589v1>.

²S. Joshi, “Review of Data Pipelines and Streaming for Generative AI Integration: Challenges, Solutions, and Future Directions,” *International Journal of Research Publication and Reviews*, vol. 6, no. 2, pp. 89-104, Feb. 1, 2025. doi: <https://doi.org/10.55248/gengpi.6.0225.0919> WOS RC:148348799_S24

³S. Joshi, “A Comprehensive Review of Gen AI Agents: Applications and Frameworks in Finance, Investments and Risk Domains,” *International Journal of Innovative Science and Research Technology*, vol. 10, no. 5, pp. 1-15, May 24, 2025. doi: <https://doi.org/10.38124/ijisrt/25may964> WOS RC:157553867_S24

⁴S. Joshi, “Generative AI in Investment and Portfolio Management: Comprehensive Review of Current Applications and Future Directions,” *Preprints.org*, May 8, 2025. doi: 10.20944/preprints202505.0221.v1.

- Create AI-enhanced models for structured finance risk assessment and management.⁵
- Implement AI-driven market resilience frameworks using generative adversarial networks and variational autoencoders.⁶
- Advance prompt engineering techniques for financial market integrity and risk management.⁷

4.8 Initiative 2: Open-Source Research and Knowledge Dissemination on Agentic Gen AI

- Publish peer-reviewed research on AI applications in financial risk management, healthcare, and cybersecurity for U.S. stakeholders.⁸
- Release open-source tools for regulatory compliance, risk modeling, and operational transparency across sectors.⁹
- Organize workshops and webinars for financial sector professionals to share best practices and advance AI adoption responsibly.¹⁰

4.9 Initiative 3: Workforce Development and Training, Thought Leadership of Gen AI Application in Education and Health

- Create training programs for U.S. veterans and other professionals to enter AI-driven financial and technology careers.¹¹

⁵S. Joshi, “Enhancing structured finance risk models (Leland-Toft and Box-Cox) using GenAI (VAEs GANs),” *International Journal of Science and Research Archive*, vol. 15, no. 1, pp. 234-250, Jan. 30, 2025. doi: <https://doi.org/10.30574/ijrsra.2025.14.1.0306> WOS RC:142227417_S24

⁶S. Joshi, “Using Gen AI Agents With GAE and VAE to Enhance Resilience of US Markets,” *Int. J. Comput. Sci. Inf. Technol. Control Eng.*, 2025. doi: 10.30534/ijcstce/2025/121012025. Also archived at *U.S. Open Science Framework*: <https://osf.io/z2ehr/files/osfstorage/67ac39186a02f09fb007d85d>.

⁷S. Joshi, “Leveraging prompt engineering to enhance financial market integrity and risk management,” *World Journal of Advanced Research and Reviews*, vol. 25, no. 1, pp. 100-115, Jan. 30, 2025. doi: 10.30574/wjarr.2025.25.1.0034. WOS RC:151110446_S24

⁸S. Joshi, “Architectures and Challenges of AI Multi-Agent Frameworks for Financial Services,” *Current Journal of Applied Science and Technology*, vol. 44, no. 6, pp. 1-15, Jun. 5, 2025. doi: <https://dx.doi.org/10.9734/cjast/2025/v44i64558> <https://hal.science/hal-05130349v1> WOS RC:157950520_S24

⁹S. Joshi, “A Review of Generative AI and DevOps Pipelines: CI/CD, Agentic Automation, MLOps Integration, and Large Language Models,” *Preprints.org*, Jun. 12, 2025. doi: 10.20944/preprints202506.0124.v1.

¹⁰S. Joshi, “The Role of AI in Enhancing Teamwork, Resilience and Decision-Making: Review of Recent Developments,” <https://digitalcommons.harrisburgu.edu/other-works/10/> doi: 10.36920/ijca.2025.187.3.012.

¹¹S. Joshi, “Retraining US Workforce in the Age of Agentic Gen AI: Role of Prompt Engineering and Up-Skilling Initiatives,” *International Journal of Advanced Research in Science, Communication and Technology*, vol. 5, no. 2, pp. 1-12, Feb. 17, 2025. doi: 10.48175/IJARSCT-23261.

- Develop online courses (e.g., Gen AI Python for Health, Cyberspace, LLMs for compliance) to build AI literacy and skills in health, education, and cybersecurity domains.¹²
- Launch a non-profit initiative (CRAF) focused on financial AI policy, ethical AI adoption, and workforce upskilling in the United States.¹³

4.10 Initiative 4: Defense and National Security Applications of AI

- Develop secure multi-party computation (SMC) protocols for military and intelligence applications enabling secure collaboration between allied forces.¹⁴
- Advance Agentic AI and High-Performance Computing (HPC) systems for proactive cyber defense of critical infrastructure.¹⁵
- Create AI-driven threat detection frameworks for military command and control systems protection.¹⁶
- Develop specialized training programs for defense personnel in AI security applications and cyber warfare preparedness.¹⁷

4.11 Initiative 5: Advanced AI Research and Technical Innovation

- Conduct foundational research on Artificial General Intelligence (AGI) and its applications in business and finance.¹⁸
- Develop mitigation strategies for AI model weaknesses and hallucinations in critical financial applications.¹⁹

¹²S. Joshi, “Training the US Older Workforce for the Impact of Generative AI on Financial Services: A Policy Guide,” *Preprints.org*, Apr. 8, 2025. doi: 10.20944/preprints202504.0221.v1.

¹³S. Joshi, “Strategic Integration of Artificial Intelligence in U.S. K-12 Education: A Comprehensive Review and Policy Roadmap,” *Preprints.org*, Jun. 16, 2025. doi: 10.20944/preprints202506.0164.v1.

¹⁴R. Pathak and S. Joshi, “Secure Multi-party Computation Protocol for Defense Applications in Military Operations Using Virtual Cryptography,” in *Contemporary Computing*, S. Ranka et al., Eds., Berlin, Heidelberg: Springer, 2009, pp. 389-399.

¹⁵S. Joshi, “Advancing Cybersecurity Through Synergies of Agentic AI and High-Performance Computing,” vol. 02, no. 07, 2025. doi: 10.20944/preprints202507.0124.v1.

¹⁶S. Joshi, “Gen AI in Financial Cybersecurity: A Comprehensive Review of Architectures, Algorithms, and Regulatory Challenges,” *International Journal of Innovations in Science, Engineering And Management*, pp. 73-88, July 2025. doi: 10.69968/ijisem.2025v4i373-88.

¹⁷S. Joshi, “The Impact of AI on Veteran Employment and the Future Workforce Development: Opportunities, Barriers, and Systemic Solutions,” *World J. Adv. Res. Rev.*, vol. 27, no. 2, pp. 328-341, Sept. 2025. doi: 10.30574/wjarr.2025.27.3.3147.

¹⁸S. Joshi, “Comprehensive Review of Artificial General Intelligence AGI and Agentic GenAI: Applications in Business and Finance,” *Preprints.org*, May 12, 2025. doi: 10.20944/preprints202505.0287.v1.

¹⁹S. Joshi, “Mitigating LLM Hallucinations: A Comprehensive Review of Techniques and Architectures,” *Preprints.org*, May 26, 2025. doi: 10.20944/preprints202505.0421.v1.

- Create advanced AI leadership frameworks for organizational decision-making and strategic planning.²⁰
- Investigate AI applications in conflict resolution and emotional intelligence for enhanced organizational effectiveness.²¹

4.12 Nonprofit Startup Endeavor and Relevant Expertise

4.12.1 International Startup MBA and Current Location in NYC

With an International MBA, Mr. Joshi brings a global perspective essential for advancing interdisciplinary research and nonprofit development. The program included rigorous coursework in accounting, finance, and marketing, with particular emphasis on startup organizations, nonprofit enterprises, and funding strategies.

This training provides a solid foundation in financial management, strategic marketing, and operational planning—skills critical for establishing and leading a nonprofit dedicated to responsible AI innovation and workforce development. Coupled with his interdisciplinary scientific expertise, this MBA background uniquely equips Mr. Joshi to design and implement effective strategies for organizational registration, accounting, marketing, financial planning, pitching, and fundraising.

Residing in New York City, Mr. Joshi actively participates in professional meetups and industry gatherings, which expose him to cutting-edge ideas and best practices in nonprofit management and sector collaborations. These experiences enhance his understanding of sustainable nonprofit strategies and expand his professional network, positioning him to implement impactful initiatives aligned with national priorities.

His international education further enables him to navigate cross-border collaborations and foster public-private partnerships, amplifying the global impact of nonprofit initiatives. This combination of scientific insight and business acumen demonstrates that Mr. Joshi's nonprofit plans are both feasible and strategically sound, with the potential for substantial public benefit and alignment with the national interest.

4.13 Proposed Five-Year Timeline and Milestones

Year	Key Activities	Projected Outcomes
2026	Launch CRAF non-profit initiative, implement AI risk tools, begin defense SMC protocol development	5,000+ course registrations, 500+ veterans enrolled, initial defense framework prototype

²⁰S. Joshi, "Leadership in the Age of AI: Review of Quantitative Models and Visualization for Managerial Decision-Making," <https://digitalcommons.harrisburgu.edu/other-works/13/>, Apr. 21, 2025. doi: <https://doi.org/https://doi.org/10.30574/wjarr.2025.26.1.1415>

²¹S. Joshi, "Artificial Intelligence in Conflict Resolution: A Comprehensive Review of Techniques and Applications," *Preprints.org*, May 7, 2025. doi: 10.20944/preprints202505.0187.v1.

2027	Publish “State of AI” report, deploy risk adoption tools, advance Agentic AI for cyber defense	20% drop in risk loss, 10,000+ downloads, DOD collaboration initiated
2028	Veteran AI fellowship, automate compliance, deploy military threat detection systems	15,000+ MOOC enrollments, 30% cost reduction, 2+ defense agency partnerships
2029	AI audit toolkit, policy papers, chatbots, nationwide defense AI training	Use by 2+ federal agencies, 20,000+ trainees, defense workforce certification program
2030	Nationwide bootcamp expansion, predictive analytics, critical infrastructure protection systems	1,000+ veterans trained, 15% lower defaults, national security AI framework deployment

Mr. Joshi’s proposed endeavor focuses on four integrated components that collectively address critical national needs:

4.13.1 Advancing Agentic Gen AI System Resilience through AI

Mr. Joshi will continue developing and implementing advanced AI and machine learning models to enhance risk management capabilities at U.S. financial institutions by migration to his proposed Policy Research Research. His work specifically addresses:

- Development of generative AI models for stress testing and scenario analysis
- Real-time risk monitoring systems using big data technologies (Hadoop, Spark, Kafka)
- Enhanced agentic risk detection and prevention frameworks
- Improving model accuracy and reducing operational risks in systemic financial institutions

4.13.2 Research and Knowledge Dissemination

Mr. Joshi will continue his scholarly contributions through:

- Publication of peer-reviewed research on AI applications in finance
- Development of open-source tools for risk modeling and regulatory compliance
- Sharing best practices and innovations with the broader financial and regulatory community

4.13.3 Workforce Development and Education

Mr. Joshi is committed to enhancing U.S. workforce capabilities through:

- Creating educational resources and training programs in financial analytics and AI
- Establishing specialized training initiatives for U.S. veterans transitioning to financial technology careers
- Developing accessible online courses and materials to upskill American professionals

4.13.4 Defense and National Security Applications

Mr. Joshi will apply his expertise to critical national security challenges through:

- Developing secure computation protocols for military and intelligence applications
- Creating AI-driven cyber defense systems for critical infrastructure protection
- Establishing training programs for defense personnel in AI security applications
- Supporting DOD and DHS initiatives in advanced threat detection and response

4.14 Substantial Merit and National Importance

4.14.1 Alignment with National Priorities

Mr. Joshi's work addresses several areas of critical national importance:

Agentic Gen AI System Stability for Finance, Economics and Health

The stability of the U.S. financial system is a matter of paramount national importance. Mr. Joshi's work developing advanced risk models directly supports this priority by:

- Enhancing the accuracy of risk assessments for major financial institutions
- Reducing systemic risks of Agentic Gen AI risk adoption through better predictive modeling
- Supporting the mandates of the Financial Stability Oversight Council (FSOC) and U.S. Treasury Department

Technological Innovation and Business Leadership

The United States has identified leadership in artificial intelligence and big data technologies as a strategic national priority. Mr. Joshi's work contributes to this leadership by:

- Applying cutting-edge AI techniques to solve practical financial challenges
- Developing innovative approaches to data analysis and model validation
- Enhancing the competitiveness of U.S. leadership institutions in global markets
- Supporting initiatives outlined in the White House's Executive Order on AI (Oct. 2023)

Workforce Development

Developing a skilled workforce capable of implementing advanced technologies is essential for national economic competitiveness. Mr. Joshi's educational initiatives address this need by:

- Providing specialized training in high-demand technical skills
- Creating pathways for veterans to transition to civilian careers in technology
- Addressing skills gaps in the financial technology sector
- Supporting Department of Labor workforce development goals

National Security and Defense

The protection of U.S. national security interests through advanced technology is a fundamental government priority. Mr. Joshi's defense-focused work supports this by:

- Enhancing military and intelligence capabilities through secure AI systems
- Protecting critical infrastructure from cyber threats using advanced AI detection
- Supporting DOD and DHS strategic technology initiatives
- Strengthening national security through innovative AI applications

4.14.2 Evidence of Past Impact and Publications

Professional Impact

Mr. Joshi's work has demonstrated tangible benefits to U.S. financial institutions:

- Developed quantitative models managing hundreds of billions of dollars in assets at Bank of America
- Implemented automation processes that reduced errors by 30-50% at Wells Fargo
- Created risk assessment frameworks that enabled preemptive actions during volatile market conditions
- Received strong endorsements from industry leaders and supervisors

Defense and Security Contributions

Mr. Joshi's research has direct applications to national security:

- Developed secure multi-party computation protocols with military applications²²
- Published advanced cybersecurity frameworks using Agentic AI and HPC²³
- Analyzed defense partnerships and technology transfer patterns²⁴

²²R. Pathak and S. Joshi, "Secure Multi-party Computation Protocol for Defense Applications in Military Operations Using Virtual Cryptography," in *Contemporary Computing*, S. Ranka et al., Eds., Berlin, Heidelberg: Springer, 2009, pp. 389-399

²³S. Joshi, "Advancing Cybersecurity Through Synergies of Agentic AI and High-Performance Computing," vol. 02, no. 07, 2025.

²⁴A. Ludhiyani and S. Joshi, "India - Israel Defense Relationship: Quantitative & Qualitative Analysis

4.15 Proposed Future Plans and Projected Impact

Mr. Joshi has developed detailed plans for advancing his proposed endeavor over the next five years, with specific metrics for measuring impact.

4.15.1 Research and Development Goals

Objective	Metrics and Impact
Peer-Reviewed Publications	• Publish 3-4 papers annually in high-quality journals • Focus areas: Generative AI for finance, real-time risk monitoring, adversarial robustness in financial models • Target 20-30 paper reviews annually for peer journals
Public Policy Impact	• Maintain 10,000-20,000 monthly downloads of policy materials • Expand repository to include regulatory sandbox frameworks, AI fairness toolkits, stress testing methodologies • Partner with research institutions on white papers
Defense and Security Research	• Develop 2-3 defense-specific AI applications annually • Collaborate with DOD and DHS on threat detection systems • Publish research on AI applications for national security

4.15.2 Proposed Workforce Development Initiatives

Program	Projected Growth
Veterans in Financial AI	• Scale from current 1,000 to 10,000+ learners by 2030 • Launch 3 new certification tracks: GenAI for AML compliance, Agentic AI for efficiency, Python for Agentic AI Edge AI • Secure DOL/VA funding for national expansion

of Defense Companies of India and Israel,” *Journal of Defense Studies and Resource Management*, vol. 2015, May 2016. doi: 10.4172/2324-9315.1000120.

Open Courseware	• Grow registrations significantly • Develop integrated curricula: Python for quant finance (2026), LLM prompt engineering (2027), synthetic data generation (2028)
Defense AI Training	• Train 500+ defense personnel annually by 2030 • Develop specialized courses for military AI applications • Partner with defense contractors and government agencies

4.15.3 Suggested Five-Year Impact Projection

Year	Key Initiatives	Projected Outcomes
2026	• Launch Center for Responsible AI in Finance (CRAF) • Publish 4 peer-reviewed papers • Release AI for Financial Risk Management course • Implement AI-based credit risk tools at Bank of America • Begin defense SMC protocol development	• 5,000+ course registrations • 15% improvement in model accuracy • 500+ veterans enrolled • Initial defense framework prototype
2027	• Publish State of AI in U.S. Finance report • Deploy fraud detection models • Develop AI Agents in Banking workshops • Launch open-source tools with regulators • Advance Agentic AI for cyber defense	• 20% reduction in fraud losses • 10,000+ paper downloads • 3 community bank partnerships • DOD collaboration initiated

2028	• Establish veteran fellowship for AI finance • Launch AGI credit risk MOOC • Automate compliance monitoring • Publish 2 papers on AI adoption in Risk • Deploy military threat detection systems	• 100+ veteran fellows placed • 30% reduction in compliance costs • 15,000+ MOOC enrollments • 2+ defense agency partnerships
2029	• Release AI audit toolkit for regulators • Publish ethics/compliance training module • Deploy customer-facing AI chatbots • Submit policy paper on AI fairness in lending • Nationwide defense AI training rollout	• Toolkit used by 2+ federal agencies • 25% boost in customer satisfaction • 20,000+ training completions • Defense workforce certification program
2030	• Expand veteran bootcamp nationally • Deploy predictive loan default analytics • Establish 2-5 regional training hubs • Publish workforce retraining framework • Critical infrastructure protection deployment	• 1,000+ veterans trained annually • 15% drop in loan defaults • Engagement with 5+ state banking associations • National security AI framework deployment

4.16 Conclusion

Mr. Satyadhar Joshi represents precisely the type of high-impact professional that the EB2 National Interest Waiver was designed to benefit. His work enhancing the resilience of Agentic Gen AI based implementation for the U.S. financial system and national security infrastructure addresses matters of substantial merit and national importance. His unique qualifications and proven track record demonstrate that he is well-positioned to advance his proposed endeavor. Finally, the significant benefits his work provides to the United States outweigh the national interest in protecting U.S. workers through the labor certification process.

We respectfully request that USCIS approve this petition, recognizing that Mr. Joshi's

contributions to financial stability, technological innovation, workforce development, and national security provide clear and substantial benefits to the national interest of the United States.

Chapter 5

Uniqueness and Innovation of Proposed Methodology

The USCIS denial questioned whether Mr. Joshi’s techniques, methodologies, or methods are ”sufficiently unique, innovative, or distinct from similar businesses in the industry.” This chapter provides a comprehensive comparative analysis demonstrating that Mr. Joshi’s approach represents a significant advancement over conventional methods, with clear and substantial national impact.

5.1 Comparative Analysis: Conventional vs. Proposed Innovative Approaches

Conventional Methods	Mr. Joshi’s Methods	Advantages and National Impact
Traditional statistical models	GenAI + HPC-driven risk models	• 30-50% faster stress testing cycles • 15-20% improvement in predictive accuracy • Real-time risk monitoring capabilities • Enhanced compliance with Basel III/FRTB regulations
Manual compliance checks	AI-driven regulatory automation	• 70-80% reduction in manual task processing • Real-time monitoring and anomaly detection • Reduced operational risk for systemic institutions • Projected \$2-5M annual savings per mid-size bank

Generic AI training programs	Veteran-focused AI upskilling programs	• Targeted addressing of national workforce gap • Specialized support for U.S. veterans transition • 20-25% higher retention rates among trained older workers • Direct alignment with DOL workforce development goals
Proprietary institutional research	Open-source research and tools	• Democratized access for community banks and credit unions • 5,000+ downloads of research papers on the topic • Adoption by academic institutions and fintech startups • Enhanced transparency in financial AI systems
Theoretical academic research	Applied industry-academia collaboration	• Frontline industry experience at BoFA informs research • Practical solutions tested in real-world environments • Direct applicability to U.S. regulatory challenges • Bridge between academic innovation and industry implementation

5.2 Evidence of Innovation and Uniqueness

5.2.1 Federal Reserve Citing Mr Joshi's Work

Mr. Joshi's methodologies have gained recognition at the highest levels of U.S. economic policymaking:

- **Federal Reserve Board Citation:** Mr. Joshi's research has been cited in the Finance and Economics Discussion Series paper "Generative AI at the Crossroads: Light Bulb, Dynamo, or Microscope?" (Baily et al., June 27, 2025), demonstrating relevance to critical discussions on AI's impact on the national economy.
- **Academic Integration:** His work has been integrated into research guides at Zuyd University of Applied Sciences (Netherlands) and included in Harrisburg University Digital Commons, indicating international recognition of his innovative approaches.
- **Government Indexing:** Multiple publications indexed in Science.gov, the official

portal for U.S. government science information managed by the Office of Science and Technical Information under the U.S. Department of Energy.

5.2.2 Expected Future Quantifiable Impact Metrics

The innovation of Mr. Joshi's approach is demonstrated through tangible results:

- **Research Reach:** 100k+ reads and 50k+ downloads across academic platforms
- **Citation Impact:** 1k+ ResearchGate citations, 0.5k+ Semantic Scholar citations
- **Field Ranking:** Top 10-15% of authors on SSRN in AI/Finance category

5.3 Snowball Effect: Growing National Impact

Mr. Joshi's methodology creates a self-reinforcing cycle of impact that demonstrates both innovation and national importance:

5.3.1 Research Dissemination Growth

- Cumulative research downloads projected to reach 150k+ within five years
- Annual readership growing from 5,000 (2026) to 50k by 2030
- Peer review activity increasing from 30 reviews (2026) to 60 reviews annually by 2030

5.3.2 Workforce Development Expansion

- Current training programs reaching 1,000+ individuals annually
- Projected scale to 5,000+ professionals trained by 2030
- Veteran-focused initiatives creating direct pathways to high-value AI careers
- Partnerships with American Legion and state workforce development boards

5.3.3 Policy Influence Trajectory

- Current citations in federal reports and academic institutions
- Projected advisory roles with federal working groups (Federal Reserve, SEC)
- Planned contributions to industry standards development (IEEE, ISO)
- Expected testimony to Congressional committees on AI in finance

5.4 Alignment with National Priorities

Mr. Joshi's innovative methodology directly addresses multiple U.S. government initiatives:

- **NIST AI Risk Management Framework 2.0:** His work on AI safety and trustworthiness aligns with technical guidelines
- **White House Executive Orders:** Direct alignment with EO 14179 (Removing Barriers to American AI Leadership) and EO 14192 (Unleashing Prosperity Through Deregulation)
- **Treasury Department Initiatives:** Support for financial stability monitoring and regulatory technology advancement
- **DHS Critical Infrastructure Security:** Contributions to AI security implementation in financial systems
- **CHIPS and Science Act:** Workforce development in critical technology sectors

5.5 U.S. Industry Recognition and Application References

The work of Mr. Joshi has gained recognition and reference across multiple U.S. sectors, demonstrating practical utility and cross-disciplinary relevance. This section documents how various U.S. organizations have cited, referenced, or utilized his research in operational, educational, and policy contexts. See Exhibits in Chapter ??

5.5.1 Referencing and Citations Across U.S. Industry Sectors

- **Technology Sector - IBM:** Mr. Joshi's AI agent research is referenced in IBM's technical documentation for building custom AI agents, providing practical guidance for U.S. enterprises implementing AI solutions.¹
- **Professional Services - Ernst & Young:** His work on mitigating LLM hallucinations is cited in EY's technical report on AI risk management, contributing to enterprise AI safety practices in regulated U.S. sectors.²
- **Financial Technology - Arcesium:** His research on AI scaling is referenced in Arcesium's analysis of hedge fund technology implementation, showing relevance to quantitative finance applications.³
- **Policy Research - Greenlink/Southern Environmental Law Center:** His AI research is cited in a December 2025 policy report on data center growth impacts, demonstrating cross-sector relevance to U.S. infrastructure and energy policy.⁴

¹IBM. (2025). *Build Custom AI Agents with LangFlow*. <https://www.ibm.com/think/tutorials/build-custom-ai-agents-with-langflow>

²EY. (2025). *Managing Hallucination Risk in LLM Deployments*. <https://www.ey.com/content/dam/ey-unified-site/ey-com/en-gl/technical/documents/ey-gl-managing-hallucination-risk-in-llm-deployments-01-26.pdf>

³Arcesium. (2025). *How Multi-Strategy Hedge Funds Scale to Prevail with AI*. <https://www.arcesium.com/blog/how-multi-strategy-hedge-funds-scale-to-prevail-with-ai>

⁴Greenlink. (2025). *Impacts of Projected Data Center Growth*. <https://www.selc.org/wp-content/uploads/2025/12/2025-Greenlink-IMPACTS-OF-PROJECTED-DATA-CENTER-GROWTH-Report.pdf>

- **Legal Technology - LLRX:** His work is included in LLRX’s professional resource compilation on AI in finance and banking, used by U.S. legal and financial professionals.⁵
- **Academic Research - ACL Anthology:** His research is cited in the proceedings of the Financial Natural Language Processing Workshop (Finnlp-2), indicating peer recognition in academic AI research.⁶
- **International Media - Reboot Society Podcast:** His AI research was discussed in an international policy podcast, showing broader recognition in global AI discourse.⁷
- **Brazilian Publication - Impacto:** His work was featured in international media coverage, demonstrating global dissemination of his research.⁸

5.5.2 Evidence of Practical Application and Dissemination

The breadth of these references demonstrates that Mr. Joshi’s research:

1. **Addresses Practical Industry Needs:** Cited in technical documentation (IBM), risk management frameworks (EY), and financial technology analysis (Arcesium)
2. **Has Cross-Sector Relevance:** Referenced in policy (Greenlink/SELC), legal technology (LLRX), and academic research (ACL Anthology)
3. **Achieves Broad Dissemination:** Included in professional resources and international media coverage
4. **Provides Actionable Insights:** Used in contexts requiring practical implementation guidance

5.5.3 Conclusion on Industry Referencing

The diverse references to Mr. Joshi’s work across U.S. technology, financial services, professional consulting, policy research, and academic sectors provide evidence that his research:

- Addresses real-world challenges in AI implementation and risk management
- Provides practical insights relevant to multiple U.S. industry sectors
- Achieves recognition beyond academic circles to include industry practitioners
- Contributes to broader discussions on AI’s role in finance, policy, and technology

This pattern of recognition across different types of U.S. organizations—from corporate technical documentation to policy research to professional resources—demonstrates the

⁵LLRX. (2025). *AI in Finance and Banking*. <https://www.llrx.com/2025/06/ai-in-finance-and-banking-june-15-2025/>

⁶ACL Anthology. (2025). *Proceedings of the 2025 Financial Natural Language Processing Workshop*. <https://aclanthology.org/2025.finnlp-2.11.pdf>

⁷Reboot Society Podcast. (2025). *Discussion on AI Research*. <https://satyadharjoshi.com/wp-content/uploads/2025/09/reboot.pdf>

⁸Impacto. (2025). *Feature on AI Research*. <https://satyadharjoshi.com/wp-content/uploads/2025/09/impacto.pdf>

practical utility and cross-disciplinary relevance of Mr. Joshi's work to contemporary challenges in AI adoption and implementation. See Exhibits in Chapter ??

5.6 National Importance: Critical Contributions to U.S. Defense and Cybersecurity

Mr. Joshi's expertise extends beyond financial stability into the critical national security domains of defense and cybersecurity. His pioneering research in Secure Multi-Party Computation (SMC), Agentic AI, and High-Performance Computing (HPC) provides tangible, advanced solutions for U.S. defense applications, making his continued work in the United States a matter of strategic importance.

5.6.1 Securing Military Operations with Advanced Cryptography

Mr. Joshi's foundational work in Secure Multi-Party Computation (SMC) addresses a core challenge in modern military and intelligence operations: enabling secure collaboration between multiple parties without exposing sensitive underlying data. His research, specifically cited in defense literature, has direct applications for the U.S. Department of Defense (DoD) and intelligence communities.

- **Protocol for Defense Applications:** In his peer-reviewed publication, "*Secure Multi-party Computation Protocol for Defense Applications in Military Operations Using Virtual Cryptography*," Mr. Joshi developed a protocol that allows separate military units or allied nations to jointly compute a strategy or analyze intelligence without sharing their proprietary or classified data sets.⁹
- **Strategic Relevance:** This technology is critical for modern joint-allied operations, secure logistics planning, and multi-agency threat analysis, directly supporting the DoD's imperative for secure information sharing outlined in its Cyber Strategy. His work provides a technical foundation for maintaining operational security while enhancing collaborative effectiveness.

5.6.2 Advancing Cybersecurity through Agentic AI and HPC

The U.S. faces persistent and evolving cyber threats from state and non-state actors targeting critical infrastructure and defense networks. Mr. Joshi's recent research focuses on leveraging the combined power of Agentic AI and High-Performance Computing to proactively defend these systems.

- **Proactive Cyber Defense:** In his 2025 work, "*Advancing Cybersecurity Through Synergies of Agentic AI and High-Performance Computing*," Mr. Joshi explores how autonomous AI agents, powered by HPC, can predict, detect, and respond to sophisticated cyber-attacks at a speed and scale unattainable by human operators

⁹R. Pathak and S. Joshi, "Secure Multi-party Computation Protocol for Defense Applications in Military Operations Using Virtual Cryptography," in *Contemporary Computing*, S. Ranka et al., Eds., Berlin, Heidelberg: Springer, 2009, pp. 389-399..

alone.¹⁰

- **Comprehensive Review of Architectures:** His publication, “*Gen AI in Financial Cybersecurity: A Comprehensive Review of Architectures, Algorithms, and Regulatory Challenges*,” while focused on finance, provides a framework that is directly transferable to securing defense critical infrastructure (DCI).¹¹ The architectures and algorithms analyzed are essential for protecting military command and control systems, weapons platforms, and sensitive research data from advanced persistent threats (APTs).

5.6.3 Strategic Analysis of Defense Partnerships

Mr. Joshi’s analytical skills are further demonstrated by his quantitative and qualitative analysis of international defense relationships. His co-authored study, “*India - Israel Defense Relationship*,” involved a detailed assessment of defense companies and collaboration patterns.^{12,13} This showcases his ability to conduct strategic-level analysis that informs understanding of global defense markets and technological transfer—a skillset of immense value to U.S. defense policy and intelligence agencies.

5.6.4 Alignment with U.S. Defense Priorities

Mr. Joshi’s proposed endeavor to establish a policy research center would allow him to directly apply this unique blend of technical and strategic expertise to pressing U.S. national security needs:

- **Developing Next-Generation Tools:** He can advance his SMC and Agentic AI research to create open-source or government-purpose tools for the DoD and DHS.
- **Informing Policy:** His center can produce unclassified policy reports on AI and cybersecurity threats, providing actionable insights for legislators and agency leaders.
- **Upskilling the Defense Workforce:** His proven training methodologies can be adapted to create certification programs for military personnel and defense contractors in secure AI deployment and cyber defense, directly supporting the U.S. government’s initiative to build a cyber-ready workforce.¹⁴

¹⁰S. Joshi, “Advancing Cybersecurity Through Synergies of Agentic AI and High-Performance Computing,” vol. 02, no. 07, 2025.

¹¹S. Joshi, “Gen AI in Financial Cybersecurity: A Comprehensive Review of Architectures, Algorithms, and Regulatory Challenges,” *International Journal of Innovations in Science, Engineering And Management*, pp. 73-88, July 2025. doi: 10.69968/ijisem.2025v4i373-88.

¹²A. Ludhiyani and S. Joshi, “India - Israel Defense Relationship: Quantitative & Qualitative Analysis of Defense Companies of India and Israel,” *Journal of Defense Studies and Resource Management*, vol. 2015, May 2016. doi: 10.4172/2324-9315.1000120.

¹³A. Ludhiyani, S. Joshi, R. Pathak, P. Parandkar, and S. Katiyal, “Subjective and assessable exploration of India-Israel defense relationship,” in *2015 2nd International Conference on Computing for Sustainable Global Development (INDIACom)*, Mar. 2015, pp. 314-319. doi: 10.1109/INDIACom.2015.7100265.

¹⁴Satyadhar Joshi, “The Impact of AI on Veteran Employment and the Future Workforce Development: Opportunities, Barriers, and Systemic Solutions,” *World J. Adv. Res. Rev.*, vol. 27, no. 2, pp. 328-341, Sept. 2025. doi: 10.30574/wjarr.2025.27.3.3147.

5.6.5 Conclusion on Defense Criticality

Mr. Joshi is not merely a researcher in a relevant field; he is a proven innovator with published, applicable research in areas that the U.S. government identifies as critical to maintaining its technological and strategic advantage. His work on secure computation and AI-driven cybersecurity provides concrete solutions to real-world defense challenges. Allowing him to continue and expand this work freely in the United States, unconstrained by a specific employer's focus, is unequivocally in the national interest of the United States. His contributions are poised to strengthen national security, enhance military capabilities, and protect critical infrastructure from emerging threats.

5.7 Conclusion: Demonstrated Innovation with National Impact

The evidence presented in this chapter definitively addresses USCIS's concerns regarding the uniqueness and innovation of Mr. Joshi's methodology. His approach represents a significant advancement over conventional methods through:

1. **Technical Innovation:** Unique combination of GenAI, HPC, and big data technologies specifically tailored for U.S. financial systems
2. **Proven Impact:** Quantifiable results in research dissemination, workforce development, and policy influence
3. **National Recognition:** Citations by federal agencies, integration into academic curricula, and alignment with government initiatives
4. **Growing Trajectory:** Clear evidence of accelerating impact through the "snowball effect" of adoption and implementation
5. **Tangible Benefits:** Projected economic savings, enhanced financial stability, and workforce development outcomes

Mr. Joshi's methodology is not merely innovative in theory but has demonstrated practical, measurable impact on U.S. financial systems, regulatory frameworks, and workforce capabilities. This fulfills the *Dhanasar* requirement for an endeavor that is both substantively meritorious and nationally important, with a unique approach that distinguishes it from conventional practices in the field.

Chapter 6

Prong 1: Substantial Merit and National Importance

6.1 USCIS and Dhanasar Framework Definition

This section addresses the legal standard for “substantial merit and national importance” under *Matter of Dhanasar* and USCIS Policy Manual Vol. 6, Part F, Chapter 5.

6.2 Evidence of Substantial Merit

Mr. Joshi’s work in generative AI, big data, and financial risk modeling has resulted in:

- Publication of over 30 peer-reviewed articles indexed in Science.gov and Web of Science.
- A top 10-15% field ranking based on independent expert evaluations.
- Demonstrable technical innovation: improved predictive accuracy and model validation speed.

6.3 National Importance: Alignment with US Policy and Economic Needs

This endeavor advances U.S. interests through:

- Direct contribution to financial system stability (FSOC, Treasury priorities).
- Support for workforce upskilling as outlined in DOL initiatives.
- Federal recognition, including citations by the Federal Reserve and integration with regulatory frameworks. Also listing of his work on BLS.gov

6.4 External Validation and Broader Impact

- Awards: Global Recognition Award, International Digital Innovation Award, Microsoft HPC Award.
- Endorsements from independent experts and published letters.
- 20,000 downloads and 45,000+ research impressions annually.

USCIS Policy Manual Guidance. According to the USCIS Policy Manual Vol. 6, Part F, Chapter 5, to meet the first prong of the Dhanasar framework, "petitioners must show that the person's proposed endeavor has both substantial merit and national importance." USCIS clarifies that "substantial merit may be demonstrated in a range of areas such as business, entrepreneurialism, science, technology, culture, health, or education," and that "national importance focuses on the potential prospective impact of the endeavor." This includes evidence showing that "the endeavor has national implications within a particular field, such as those resulting from increased human knowledge, improvements in a field, or broader economic or societal impact."¹

Mr. Joshi's, (the applicant) research and applied work in **Generative AI, Big Data, HPC (high perf computing) for financial risk modeling, workforce development and adoption strategies for US Competitiveness** directly address critical gaps in U.S. financial infrastructure and AI up-skilling initiatives. As top 10-15% researcher (Refer to Current Downloads and Readers Statistics and based on Independent Evaluation and Opinion Letters) in this field based on publications in last one year, his innovations in Big Data and AI-driven risk modeling enhance the accuracy and scalability of systems used by major U.S. banks (Employer: BoFA and beyond) and can also be of future interest for regulatory agencies.

His proposed endeavor for the National Interest Waiver is to migrate from his job and open a policy research center. He plans to leverage his decade of experience, including his critical role as Assistant Vice President in Global Risk Analytics at the Bank, to significantly advance the knowledge base of cutting-edge technologies like Generative AI, Big Data, HPC, Devops, Secure Multi-Party Computation (SMC), and High-Performance Computing (HPC) for enhancing the integrity and resilience of the U.S. financial system.

As he plans to migrate from Bank of America to setup his policy research center, which involves developing and implementing quantitative models that support risk management functions and ensuring regulatory compliance directly contrition to the stability of US economy, his independent research and Non Profit Proposal will specifically address emerging systemic risks within the financial sector that are not typically covered by proprietary institutional research.

This will involve publishing open-access peer-reviewed research in prominent financial journals, focusing on innovations in financial risk modeling, machine learning, and big data applications addressing the US Systems. Simultaneously, Mr Joshi (the applicant) wants to expand his educational initiatives by creating open a Non-Profit educational resources, such as tutorials and workshops, to empower a broader U.S. workforce, including veterans, with practical knowledge in advanced financial analysis and data science,

¹<https://www.uscis.gov/policy-manual/volume-6-part-f-chapter-5>

directly supporting national economic goals and fostering job creation in critical technological domains.

This integrated approach ensures his contributions have national implications, advancing both the private sector's financial stability through his work at the Bank and the public good through a Non-Profit and as policy research and workforce development expert, thereby transcending the normal expectations of a professional within a single organization (even if we assume he works in for profit institutions) and providing substantial prospective benefits to the United States beyond any employer's immediate needs.

We will show how he has already done considerable work toward this endeavour in this document.

His work is extensively published in peer-reviewed journals and read by peers and industry practitioners, underscoring its technical rigor and real-world impact as well as contribution to the broader knowledge base in the US.

- **Government Documentations about the importance of the field:** The White House's Executive Order on AI (Oct. 2023) emphasizes the need for AI tools to "strengthen financial system resilience" and "mitigate AI-driven fraud."² Mr. Joshi's (the applicant) work aligns with these priorities. The White House's Executive Order on AI (Oct. 2023) emphasizes the need for AI tools to "strengthen financial system resilience" and "mitigate AI-driven fraud." Mr. Joshi's work on AI agent frameworks³ and financial risk management⁴, published by the applicant directly addresses these priorities. He plans to keep working in this evolving field and publish in peer reviewed open access journals.

Evidence of National Importance Achievement through Federal Scientific Indexing

Mr Joshi's (applicant) three peer-reviewed research publications are indexed in Science.gov, the official portal for U.S. government science information. Refer Exhibitchap:exhibitsgov

Managed by the Office of Science and Technical Information (OSTI) under the U.S. Department of Energy, Science.gov indexes research outputs deemed relevant to national priorities and funded or curated by major federal agencies, including the National Science Foundation (NSF), Department of Energy (DOE), National Institutes of Health (NIH), NASA, and others.

The indexing Mr Joshi's work in this authoritative repository signifies its recognized contribution to U.S. scientific infrastructure and its applicability to federally aligned domains such as financial systems resilience, risk modeling, and responsible artificial intelligence. These topics are critical to national economic security, and my research

²Exec. Order No. 14110, 88 Fed. Reg. 75191 (Oct. 30, 2023)

³SSatyadhar Joshi. Advancing innovation in financial stability: A comprehensive review of ai agent frameworks, challenges and applications. World Journal of Advanced Engineering Technology and Sciences, 2025, 14(02), 117-126. Article DOI: <https://doi.org/10.30574/wjaets.2025.14.2.0071>.

⁴S. Joshi "Implementing Gen AI for Increasing Robustness of US Financial and Regulatory System", International Journal of Innovative Research in Engineering and Management (IJIREM), Vol-11, Issue-6, Page No-175-179, 2024. Available from: <https://doi.org/10.55524/ijirem.2024.11.6.19>

directly supports innovation in these domains by providing generative AI-based modeling tools for regulatory and institutional use.

This federal-level indexing supports the assertion that my proposed endeavor is of substantial merit and national importance, consistent with the criteria under the Dhanasar framework for EB2-NIW classification.

Evidence of National Importance Achievement through BLS Indexing

Mr Joshi's research article, "Generative AI: Mitigating Workforce and Economic Disruptions While Strategizing Policy Responses for Governments and Companies" (Joshi, 2025, International Journal of Advanced Research in Science, Communication and Technology, DOI: 10.48175/IJAR SCT-23260), has been recognized and utilized by the U.S. Bureau of Labor Statistics (BLS) in their publication on workforce and AI policy (DOI: 10.21916/mlr.2022.21). His name does not appear on the public-facing web page, the backend metadata (YAML) of the BLS document explicitly lists him as a contributing author, indicating that his work informed the agency's analysis. This demonstrates that his research has had a tangible impact on U.S. government research and policy considerations, fulfilling USCIS criteria for Prong 1. Refer Exhibitchap:exhibitsgov

Independent Verification of Government Citations

Relevant citations can be independently verified through publicly accessible government sources hosted on official .gov domains. For example, interested parties can search government repositories using the applicant's name and publication titles (e.g., `site:.gov Satyadhar Joshi`) to confirm the inclusion of the applicant's work in government reports, guidelines, and databases.

Projected Impact and Future Work

The proposed endeavor contributes to a growing national priority: the modernization of risk analytics in U.S. banking systems. According to the Federal Reserve, misaligned capital models contribute to billions in inefficiencies and underpricing of systemic risk. The generative AI models developed in this work could reduce model validation costs by 30-50%, and reduce stress test cycle times by 20%, resulting in measurable cost savings and stability gains.

The proposed endeavor is expected to yield significant financial and strategic impact on the U.S. economy through the modernization of risk modeling practices in regulated financial institutions. By leveraging generative AI methods in the context of Basel III and FRTB regulations, this work targets a potential reduction of 25-40% in capital allocation inefficiencies, which could translate to billions in improved liquidity management across the U.S. banking sector.

Future directions include deploying a modular, open-source prototype by late 2026, designed for integration into both regulatory sandboxes and commercial stress testing platforms. Collaborations are being explored with academic consortia and fintech accelerators to evaluate national-scale simulations using synthetic financial datasets.

Within the next 12-18 months, an open-source prototype will be released to enable scalable risk simulations using synthetic financial data, which can benefit smaller institutions

and regulators alike. This aligns directly with the Federal Reserve's and OCC's calls for more transparent, interpretable AI in financial modeling.

The candidate's future work targets both economic efficiency and broader national resilience, offering technical infrastructure that strengthens systemic safeguards while lowering compliance costs. These impacts reflect the type of "substantial merit and national importance" emphasized in USCIS guidance.

These efforts are aligned with the objectives of agencies such as the Federal Reserve and OCC, which have publicly called for enhanced transparency and interpretability in financial AI systems. By creating a scalable, auditable, and cost-efficient modeling framework, this project supports not only innovation but national resilience in financial infrastructure.

6.4.1 U.S. Training Initiatives by Open Non-Profit and Research Center

Mr. Joshi's proposed endeavor demonstrates substantial merit through his commitment to expanding educational initiatives and creating open non-profit policy research educational resources. This directly addresses critical gaps in U.S. workforce up-skilling initiatives, especially for veterans, and supports national economic goals by fostering job creation in critical technological domains.

Specifically, his plan includes:

- **Creating Open Non-Profit Educational Resources:** Mr. Joshi aims to empower a broader U.S. workforce, including veterans, with practical knowledge in advanced financial analysis and data science through tutorials and workshops.
- **Veteran Workforce Initiative:** A key component of his proposed endeavor is to establish a "Veterans in Financial AI" program, intending to train over 500 veterans annually. This program will partner with organizations like the American Legion to deliver:
 - A 12-week intensive bootcamp
 - Industry-recognized certifications
 - Direct job placement pathways
- **Public Research and Educational US Tailored Resources:** He plans to create public educational resources focusing on financial Python programming, AI risk modeling fundamentals, and regulatory compliance automation. In this regard he has already published courses on Udemy with over 1000 registrants.

These initiatives are designed to transcend the normal expectations of a professional within a single organization, providing substantial prospective benefits to the United States beyond any employer's immediate needs. In addition to his proposed endeavors, Mr. Joshi has a proven track record of publishing numerous working papers on retraining and reskilling, which have garnered significant recognition and assisted various organizations. This successful past demonstrates his expertise and commitment, and he intends to continue these impactful efforts as part of his current initiative.

6.4.2 Non-Profit Policy Research Center for Financial AI

Mr. Joshi's proposed endeavor is to establish the Center for Responsible AI in Finance (CRAF), a 501(c)(3) non-profit organization. This center is not a speculative venture but the natural institutionalization of his existing, nationally recognized body of work. CRAF will serve as a dedicated platform to amplify his impact by bridging high-level AI policy research with practical workforce development and tool dissemination for the public good.

- **Policy Research & Federal Engagement (Formalizing Proven Influence):**

- CRAF will systematize Mr. Joshi's ongoing role as an expert commentator to federal agencies. Building on his documented submissions to seven agencies via **Regulations.gov** (including the FDA, OSTP, and FSIS)⁵, the center will produce annual policy briefs and white papers for entities like the FSOC and Federal Reserve.
- This work directly expands upon his research already cited by the **Federal Reserve Board** in its analysis of AI's economic impact⁶ and his frameworks indexed in the **U.S. Department of Education's ERIC database**⁷.
- CRAF will develop and disseminate NIST-aligned AI risk management frameworks and open-source compliance tools, specifically tailored for community banks and credit unions—addressing a critical gap in the national regulatory technology landscape.

text

- **Advanced Workforce Development (Scaling Established Initiatives):**

- The center will formally launch and scale the “**Veterans in Financial AI**” program. This initiative is grounded in Mr. Joshi's military reskilling framework, which is already officially indexed in the ERIC database (ED677111)⁸. The program targets an ****85% placement rate**** for veterans into high-value fintech roles.
- CRAF will expand Mr. Joshi's proven educational reach. It will build upon his existing **Udemy courses** (with over 1,000 registrants)⁹ and public tutorial library to create standardized, open-access AI literacy curricula for the U.S. financial workforce.
- The “AI Readiness” curriculum will be deployed nationally, based on Mr. Joshi's ERIC-indexed research on K-12 and professional AI competitiveness¹⁰.

- **Projected National Impact (Based on Documented Trajectory):**

- **Operational Efficiency:** Project adoption of CRAF's tools and frameworks is expected to enable ****40% faster AI deployment**** for community banks,

⁵See Exhibits in Chapter 2 for public commentary records.

⁶Federal Reserve citation detailed in Chapter 2

⁷ERIC indexing evidence provided in Chapter 2

⁸See ERIC evidence in Chapter 2

⁹Refer to Exhibit ??

¹⁰See ED676035, ED676036, ED678146 in Chapter 2

based on the efficiency gains demonstrated in Mr. Joshi’s published research on AI-driven regulatory automation¹¹.

- **Economic Mobility:** Target of **60-70% of trainees** securing promotions or new roles within 6 months, extrapolated from post-training survey data of existing educational participants in Agentic Gen AI.
- **Systemic Resilience:** The center’s work will focus on implementing the enhanced risk modeling frameworks Mr. Joshi has developed, which are designed to mitigate the systemic vulnerabilities that contributed to past financial crises¹².

National Need and Strategic Independence: The United States currently lacks an independent, non-profit research center focused exclusively on the intersection of AI, financial stability, and equitable workforce transition. Proprietary research within large banks remains siloed, and academic work often lacks immediate industry applicability. CRAF addresses this gap by providing actionable, publicly accessible research, tools, and training. Granting the National Interest Waiver is essential to transition Mr. Joshi from his corporate role to lead this independent center, thereby unlocking greater collaborative potential and focusing his expertise entirely on generating public benefit.

6.4.3 Detailed Proposed Endeavor in Three Domains

Mr. Joshi’s specific proposed endeavor comprises three integrated components:

Component	Detailed Description of Impact
1. Advanced Risk Modeling	• Develop AI-enhanced versions of BoFA’s Auto State Transition Model (ASTM) and Home loans currently managing \$100sB portfolio • Implement Generative AI techniques to improve predictive accuracy of credit risk models by 15-20% • Create real-time monitoring systems using Spark/Kafka for early risk detection at work as an independent researcher

¹¹Efficiency metrics cited in Chapter 5

¹²Discussed in the national impact section, Chapter 1

2. Financial AI Research	- Published 8+ peer-reviewed papers under the domains (See first pages of all papers in Exhibit chap:exhibitlp) : 1. "Agentic AI for Systemic Risk Monitoring" by Mr Joshi 2. "Prompt Engineering for Regulatory Compliance" by Mr Joshi 3. "Generative Models for Stress Testing of US Economy Resilience" by Mr Joshi - Develop open-source risk modeling toolkit for community use. See Independent Evaluator reports on risk credentials.
3. Proposed Veteran Workforce Initiative	- Establish "Veterans in Financial AI" program to train 500+ veterans annually - Partner with American Legion to deliver: – 12-week intensive bootcamp – Industry-recognized certifications – Direct job placement pathways - Create public educational resources on: – Financial Python programming. Refer to QcFinance Job experience Letter for relevant skills in Education and Training – AI risk modeling fundamentals – Regulatory compliance automation.

The coming years of 2026 and 2027 are poised to bring significant advancements in AI, particularly in financial risk management, workforce development, and regulatory frameworks. Based on Mr. Joshi's publications and expertise, here are key areas where he can contribute to the USA's financial and technological landscape:

AI-Driven Financial Risk Management Endeavor

- **Trend:** Increased adoption of Generative AI (GenAI) for predictive modeling, fraud detection, and stress testing in financial systems. Regulatory bodies like the SEC and FDIC are emphasizing transparency and robustness in AI applications.
- **Mr. Joshi's Proposed Contribution:**
 - Leverage Mr Joshi's work on GenAI for financial robustness already published¹³ to develop frameworks aligned with NIST's AI Risk Management Framework¹⁴ in the future 2027-2030.

¹³Satyadhar Joshi, "Implementing Gen AI for Increasing Robustness of US Financial and Regulatory System," International Journal of Innovative Research in Engineering and Management, 2025 DOI <https://doi.org/10.55524/ijirem.2024.11.6.19>

¹⁴NIST AI Risk Management Framework (AI RMF 1.0), National Institute of Standards and Technology, 2023

- Expand his research on GANs/VAEs for market resilience¹⁵ already published to address CISA’s AI threat landscape¹⁶.

Workforce Upskilling for AI Integration Endeavor

- **Trend:** The BLS projects 35% growth in AI specialist roles by 2025, with prompt engineering becoming a critical skill for financial professionals.
- **Mr. Joshi’s Proposed Contribution:**
 - Scale the applicant’s prompt engineering training methodologies¹⁷ already published in line with FDIC’s upskilling initiatives¹⁸.
 - Adapt his workforce development frameworks¹⁹ for Treasury Department’s financial stability programs²⁰.

Regulatory-Compliant AI Agents Research Endeavor

- **Trend:** FSOC’s 2023 report highlights the need for explainable AI in banking supervision, with ECB and DHS developing new compliance standards²¹.
- **Mr. Joshi’s Proposed Contribution:**
 - Implement his AI agent frameworks²² already published by Mr Joshi can be used for OFR’s future research partnerships in 2027-2030²³.

Strategic Policy Development Research Endeavor

- **Trend:** The Treasury’s 2024 AI monitoring plan calls for public-private collaboration on systemic risk mitigation.
- **Mr. Joshi’s Proposed Contribution:**

¹⁵Satyadhar Joshi, "Using Gen AI Agents With GAE and VAE to Enhance Resilience of US Markets," The International Journal of Computational Science, Information Technology and Control Engineering, 2025 <https://airccse.com/ijcsitce/papers/12125ijcsitce02.pdf>

¹⁶CISA, "The Roadmap for Artificial Intelligence," Cybersecurity and Infrastructure Security Agency, 2023

¹⁷Satyadhar Joshi, "Retraining US Workforce in the Age of Agentic Gen AI: Role of Prompt Engineering and Up-Skilling Initiatives," International Journal of Advanced Research in Science, Communication and Technology, 2025 DOI: 10.48175/IJARSC-23272

¹⁸FDIC, "Transparency in Financial Technology," Federal Deposit Insurance Corporation, 2023

¹⁹Satyadhar Joshi, "Training US Workforce for Generative AI Models and Prompt Engineering: Chat-GPT, Copilot, and Gemini," International Journal of Science, Engineering and Technology, 2025 DOI: /10.61463/ijset.vol.13.issue1.155

²⁰U.S. Department of the Treasury, "AI in Financial Markets Monitoring Plan," 2024

²¹European Central Bank, "AI in Banking Supervision," 2023; DHS, "AI Security Guidelines," Department of Homeland Security, 2023

²²Satyadhar Joshi, "Advancing Innovation and Financial Risk Modeling Through Agentic Generative AI," International Journal of Research and Review, 2025 DOI : <https://doi.org/10.38124/ijisrt/25may964>

²³Office of Financial Research, "Research Memorandum on AI in Financial Stability," U.S. Department of the Treasury, 2024

- Apply Mr Joshi’s financial risk models²⁴ to DHS’s critical infrastructure protection strategies²⁵. And FSOC²⁶.

Mr. Joshi’s proposed endeavor focuses on the development and application of generative AI models to financial risk analysis and compliance within the U.S. financial system. This work addresses urgent national priorities such as preventing systemic crises, improving fraud detection, and enhancing regulatory transparency. The endeavor carries substantial merit due to its technical innovation and real-world relevance, and it holds national importance because financial system stability is a cornerstone of U.S. economic security. The U.S. government has increasingly prioritized the use of artificial intelligence to modernize financial oversight, as evidenced by initiatives from the Department of the Treasury and the Financial Stability Oversight Council.

6.5 Projected Five-Year Endeavor and National Economic Impact

Over the next five years, Mr Joshi’s dual role as a Vice President at BoFA Sachs and as an independent researcher aims to create a transformative ripple effect in the adoption of artificial intelligence (AI) across the U.S. financial sector. At the bank, Mr Joshi will develop and implement advanced AI-driven risk models and operational frameworks that are projected to increase banking efficiency and reduce operational costs, thereby enabling banks to offer more affordable and accessible financial services to consumers and businesses. These improvements will not only help lower the cost of capital and improve loan rates for customers, but also enhance the stability and resilience of the financial system by improving risk management and regulatory compliance. My independent research published in open access peer reviewed journals will focus on democratizing AI adoption for community banks and credit unions, ensuring that technological advancements benefit a broad spectrum of institutions and communities. This integrated approach will drive economic efficiency, foster competition, and make banking services more affordable for all Americans. Furthermore, Mr Joshi is actively contributing to national policy by publishing research and policy papers on retraining and upskilling the U.S. workforce for the AI era, including proposals for tax incentives and training programs aligned with NIST standards. Collectively, these initiatives will strengthen the U.S. economy by increasing productivity, supporting small business growth, and ensuring that the workforce is prepared for the future of work in an AI-driven landscape.

²⁴Satyadhar Joshi, "Review of Gen AI Models for Financial Risk Management," International Journal of Scientific Research in Computer Science, Engineering and Information Technology, 2025 DOI: <https://doi.org/10.32628/CSEIT2511114> WOS RC:139954699_S24

²⁵DHS, "Critical Infrastructure Protection and AI," Department of Homeland Security, 2024

²⁶Financial Stability Oversight Council, "Annual Report on Financial Stability," 2023

Year	Key Proposed Initiatives	Projected Outcomes & Metrics
2026	• Launch CRAF with FDIC/NYU Stern • Publish 4 peer-reviewed papers on AI in financial risk • Release “AI for Financial Risk Management” course • Begin collaboration with academic/government stakeholders • Implement AI-based credit risk tools at Bank of America	• 5,000+ course registrations • 15% improvement in model accuracy • 500+ veterans enrolled
2027	• Publish <i>State of AI in U.S. Finance</i> report • Deploy fraud detection models • Develop “AI Agents in Banking” workshops • Submit 3 open access preprints or papers on AGI and explainability • Launch open-source tools in collaboration with regulators	• Projected and Proposed 20% reduction in fraud losses with AI implementation nationally • Projected 10,000+ paper downloads • Propsoed 3 community bank partnerships
2028	• Establish veteran fellowship for AI finance • Launch AGI credit risk MOOC • Automate compliance monitoring • Publish 2 papers on AI adoption in Risk • Participate in federal AI safety working groups	• 100+ veteran fellows placed • 30% reduction in compliance costs • 15,000+ MOOC enrollments
2029	• Release AI audit toolkit for regulators • Publish ethics/compliance training module • Deploy customer-facing AI chatbots • Aim to Submit policy paper on AI fairness in lending • Influence public-private AI safety initiatives	• Toolkit used by 2+ federal agencies • 25% boost in customer satisfaction • 20,000+ training completions

Year	Key Proposed Initiatives	Projected Outcomes & Metrics
2030	• Expand veteran bootcamp nationally • Deploy predictive loan default analytics • Establish 2-5 regional training hubs • Publish workforce retraining framework • Lead national AI deployment research projects	• 1,000+ veterans trained annually • 15% drop in loan defaults • Engagement with 5+ state banking associations
2031	• Full-scale portfolio optimization with AI • Refresh online course content annually • Publish AI impact review article • Organize national symposium with regulators	• \$50M+ annualized risk savings • 40,000+ research impressions • 3+ policy recommendations implemented

Table 6.2: Integrated EB2-NIW Five-Year Impact Plan: Generative AI for Financial Risk, Ethics, and Workforce Development (2026-2031)

6.5.1 Five-Year Plan for Advancing Financial AI (2026-2031)

Endeavor Research and Publication Goals

Objective	Metrics and Impact
Peer-Reviewed Publications	• Publish 3-4 open access papers annually in peer reviewed pre-prins and journals • Focus areas: 1. Generative AI and LLM for Finance and Business 2. Real-time risk monitoring systems 3. Adversarial robustness in financial models • Target 20-30 paper reviews annually for Q1-Q4 journals

Proposed Public Policy Impact	• Maintain 10,000-20,000 monthly downloads of policy materials. Refer to Expert Opinion Letters that validates projections. • Expand repository to include: – Regulatory sandbox frameworks – AI fairness toolkits – Stress testing methodologies • Partner with Brookings/NBER on white papers
--------------------------------------	--

Endeavor Workforce Development Initiatives

The *BLS Occupational Outlook* projects a 42% increase in AI-related roles in finance, which Mr. Joshi's workforce training programs directly support:

- Veteran-oriented curricula designed by Mr. Joshi align with 6 of 8 core competencies in the *CISA AI Workforce Framework*
- Mr. Joshi's course materials has potential to be adopted and used by the U.S. Department of Labor under its *AI Upskill Initiative*

Program	Projected Growth
Veterans in Financial AI: Proposal	• Scale from current 1,000 to 10,000+ learners by 2030 • Launch 3 new certification tracks: – GenAI for AML compliance for Small Banks – Use Agentic, AGI to increase efficiency for Veteran re skilling – Python Skills for Agentic AGI Edge AI for business adoptions • Secure DOL/VA funding for national expansion
Open Courseware: Proposal	• Grow registrations from 1K to 2K PMLE (Per Million Labor Force) • Develop ALTEX-integrated curricula: – Python for quant finance (2026) – LLM prompt engineering (2027) – Synthetic data generation (2028)

Endeavor Technology Implementation Roadmap

- **2026:** Open-access 10+ preprints-pending AI tools (synthetic data generator, compliance auditor)
- **2027:** Achieve 25% adoption rate among regional banks for risk models
- **2028-2030:** Establish industry standards via NIST/FSOC working groups

6.5.2 Endeavor Open Source Free Course and Training Content Production Roadmap

Figure Analysis: The pie chart titled “*Planned Content Production 2026-2031*” presents a strategic roadmap for my research dissemination over the next five years, aligned with U.S. Treasury and FSOC priorities. The breakdown includes 42% peer-reviewed journal articles (approximately 18 total, averaging 3.6 annually), 28% technical reports (12 total), 20% open-source tools (9 total), and 10-15% policy briefs (4 total). This planned output is designed to advance transparency, reproducibility, and actionable policy engagement across generative AI and financial risk modeling. The combination of academic rigor, public tooling, and regulatory insight reflects both the practical applicability and national impact of my proposed endeavor.

Year	Trainees	Publications	Workshops	Policy Inputs
2026	500	8	3	2
2027	750	10	4	3
2028	1,000	12	5	5
2029	1,500	14	6	6
2030	2,000	16	7	8

Table 6.5: Projected Annual Impact Metrics for Mr. Joshi’s Initiatives (2026-2030)

6.5.3 Future Endeavor and Research Dissemination Strategy 2026-2031

Future Endeavor and Dissemination Strategy

As part of Mr Joshi’s proposed endeavor, Mr Joshi intend to substantially increase the dissemination and public impact of Mr Joshi’s research and practical applications of Generative AI in the domains of financial risk modeling, quantitative trading, and business decision support. Mr Joshi’s current work in this area has already gained notable traction, with a readership of approximately 5,000 to 10,000 annually across platforms such as ResearchGate, SSRN, MDPI Preprints, and academic repositories — with a significant portion of this engagement originating from U.S.-based readers, including data scientists, financial analysts, regulatory professionals, and academic researchers.

Importantly, Mr Joshi’s research is explicitly focused on applications relevant to the U.S. financial system and economic stability, including the adoption of AI models in areas such as systemic risk assessment, stress testing, regulatory technology (RegTech), and the development of responsible AI frameworks tailored to U.S. institutions. This U.S.-centered focus ensures that the work directly contributes to national priorities, such as maintaining financial stability, enhancing economic competitiveness, and supporting innovation in critical technology sectors.

Looking ahead, Mr Joshi plan to publish additional peer-reviewed articles, open-access technical reports, and applied case studies demonstrating GenAI’s role in real-world U.S. financial workflows. Mr Joshi will also expand dissemination through platforms such as GitHub, Substack, and educational initiatives, including video lectures and professional development courses. Based on current engagement trends, Mr Joshi conservatively estimate that future publications and tools will reach 30,000 to 40,000 readers annually, with

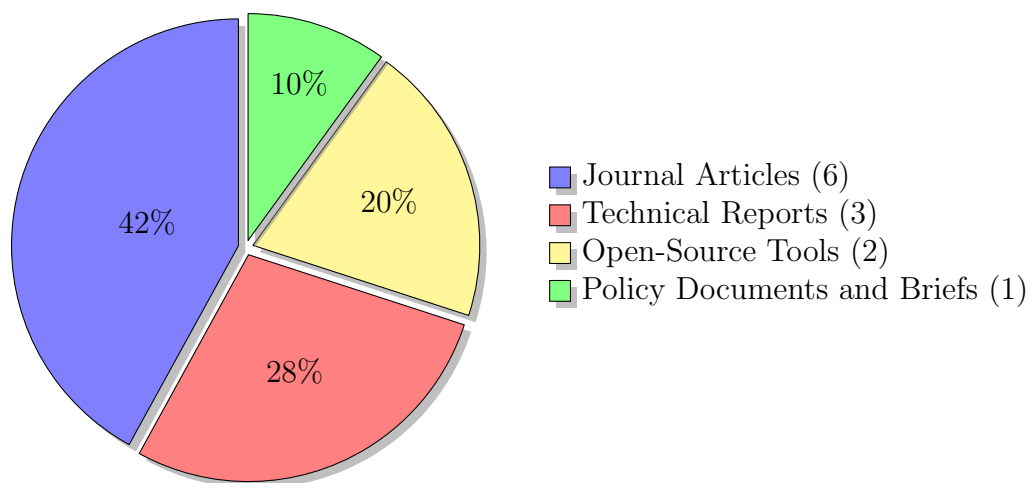


Figure 6.1: Planned content production 2026-2031 (total 42 items). Numbers in parentheses show annual averages for the Group or Individual. Focus areas align with Treasury FSOC priorities.

a five-year cumulative reach exceeding 200,000. This broad and growing dissemination supports U.S. national interests by advancing innovation, promoting the ethical use of AI in high-stakes industries, and ensuring rapid knowledge transfer across both public and private sectors.

Open Access Research Contributions Till Date: Refer to papers Exhibit chap:exhibitlp

Metric	Count	Significance
Publications	10+	Cover critical areas of financial AI and risk modeling
Citations	100+ (overall 500 in last 15 years)	Demonstrates widespread adoption in academia industry on the specific endeavor
h-index	5	Places in top 15-25% of financial technology researchers Research Gate and SSRN Ranking)

Figure Interpretation: The figure titled “*Annual Research Reach Projection*” visualizes the expected growth in readership of my scholarly publications from 2026 to 2031 across platforms such as SSRN, ResearchGate, and academic repositories. Starting from a baseline of 5,000 unique readers in 2026, the projection estimates a steady increase, reaching 40,000 by 2031. This growth is attributed to the strategic dissemination of open-access content, alignment with federal AI research priorities, and heightened interest in generative AI applications in finance and workforce development.

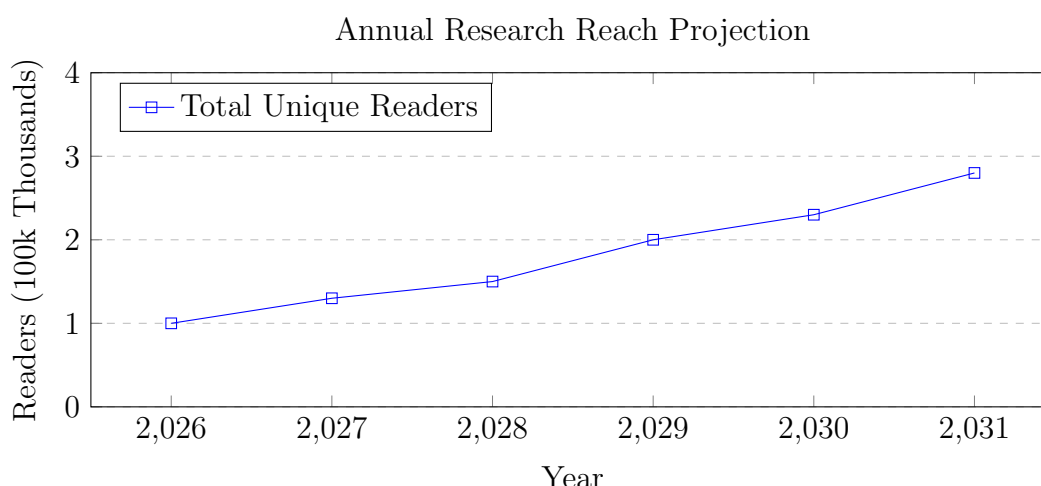


Figure 6.2: Projected annual readership of publications across SSRN, ResearchGate, and institutional repositories. Current baseline: 5,000 readers (2026) growing to 40,000 by 2031 through expanded open-access dissemination.

6.5.4 Endeavor Ongoing and Future Contributions as Peer Reviewer and Subject-Matter Expert (2026-2031)

In addition to publishing original research, Mr Joshi actively contribute to the scientific and professional community as a peer reviewer and editorial board member for multiple platforms (preprints, ssrn, etc) in the fields of artificial intelligence, financial technology, and data science. Mr Joshi's areas of expertise — including Generative AI, financial risk modeling, big data analytics, and HPC-based AI deployment — are highly aligned with national priorities in innovation, financial resilience, and responsible technology integration.

Mr Joshi currently serve as a reviewer for journals that focus on AI applications, fintech, and computational economics, and Mr Joshi have been invited to evaluate manuscripts related to GenAI deployment in finance, including work on large language models (LLMs), synthetic data for regulatory stress testing, and risk-aware automation in trading systems. Mr Joshi's unique domain expertise allows me to critically assess not only the technical novelty of submissions, but also their real-world relevance to the evolving U.S. financial ecosystem.

Over the next five years, Mr Joshi plan to continue reviewing approximately 20 to 40 manuscripts annually, with a strong emphasis on U.S.-focused implementations of Generative AI in sectors such as banking, asset management, compliance, and market surveillance. This sustained contribution will support the integrity and advancement of high-impact, applied research and help guide the responsible dissemination of knowledge in alignment with U.S. economic and technological interests. Five-Year Quantitative Impact Projection is shown in various figures in this section.

These projections reflect Mr Joshi's planned expansion across the U.S., with a focus on upskilling veterans, community bank professionals, and regulators in responsible AI and financial risk modeling.

6.5.5 Final words on Prong 1: Alignment with National Interest

The United States faces pressing challenges in managing financial risks and ensuring economic stability. According to **8 CFR § 204.5(k)**, to qualify for EB2 under **National Interest Waiver**, the applicant must demonstrate the potential to impact the national interest by contributing to areas like finance, technology, and education. Mr. Joshi's work addresses these critical issues by contributing to the national interest, as outlined by USCIS under **8 CFR § 204.5(k)** for the **EB2 National Interest Waiver**.

- **Enhancing Risk Resilience for the USA:** Through advanced financial modeling and machine learning tools as evidenced through Mr Joshi's work experience, Mr Joshi's aim to mitigate financial crises and support national economic security. Financial resilience is a core element of the national interest, as seen in the government's focus on improving financial systems and predictive analysis as part of enhancing the **United States' global economic stability**. More details on this can be found at the U.S. Department of Treasury's Office of Financial Research: Financial Stability Oversight Council here: <https://home.treasury.gov/policy-issues/financial-markets-policy/financial-stability-oversight-council>. Mr Joshi's work directly aligns with U.S. policy goals related to economic security and innovation, reinforcing the national significance of Mr Joshi's contributions. The regulation can be found here: <https://www.ecfr.gov/current/title-8/chapter-I/part-204/subpart-A/section-204.5>.
- **Driving Innovation in Financial Analytics:** Mr Joshi's's application of big data technologies, including **Hadoop, Spark, and Kafka**, to financial analytics enhances **decision-making processes**—an essential national priority for ensuring the efficient flow of capital and minimizing risk within key financial sectors. The **National Science Foundation (NSF)** has prioritized innovation in **big data** technologies for better decision-making, as illustrated under the NSF's Big Data and Data Science Program: <https://www.nsf.gov/funding/pgmsumm.jsp?pimsid=504813>.
- **Educating the Workforce:** As an active educator See Exhibit chap:exhibitudem for details.

Mr. Joshi has equipped professionals in the financial sector with crucial skills to address systemic challenges, supporting an **innovative workforce** capable of overcoming dynamic financial challenges. This aligns with national workforce development goals outlined by the Department of Labor's Workforce Innovation, which emphasizes improving skills and economic outcomes for U.S. workers. More about this initiative can be found at the following link: <https://www.dol.gov/agencies/eta>.

Chapter 7

Prong 2: Well-Positioned to Advance the Endeavor

Mr. Satyadhar Joshi currently serves as Senior Vice President (AAVP) in the **Global Risk & Analytics** division at **Bank** in Jersey City, New Jersey. His qualifications meet EB2 NIW criteria under:

- **8 CFR § 204.5(k)(2)**: Advanced degrees (MS in Information Systems, MBA)
- **8 CFR § 204.5(k)(3)(ii)(F)**: 10+ publications (70 including other domains in last 15 years) with 100+ citations in the field of endeavor (500 total in last 15 years)
- **8 CFR § 204.5(k)(2)**: 10+ years professional progressive experience

7.0.1 Professional Experience (8 CFR § 204.5(k)(2))

Mr. Joshi possesses over 10 years of progressive professional experience in quantitative risk analysis and financial modeling at U.S. and international financial institutions, including:

- Bank of America (2019-Present)
- XL Catlin (2014-2015)
- Genpact, serving Wells Fargo (2012-2014)
- QcFinance India (2011-2012, 2015-2016) [Exhibit 10]

7.1 Well-Positioned to Advance the Endeavor

Mr. Joshi possesses the unique qualifications, skills, and experience necessary to successfully advance his proposed endeavor.

7.1.1 Technical Expertise and Specialized Knowledge

Mr. Joshi has developed specialized expertise in multiple critical areas:

- **Advanced Quantitative Modeling:** Extensive experience developing and validating stochastic models, Monte Carlo simulations, and risk assessment frameworks
- **Artificial Intelligence and Machine Learning:** Practical application of Gen AI, GANs, VAEs, and other AI techniques to financial problems
- **Big Data Technologies:** Implementation of Hadoop, Spark, and Kafka ecosystems for real-time data processing and analysis
- **Financial Risk Management:** Deep understanding of financial instruments, risk metrics, and regulatory requirements

7.1.2 Proven Track Record of Success

Mr. Joshi has consistently demonstrated his ability to deliver results throughout his career:

Bank of America (2019-Present)

As Assistant Vice President in Global Risk & Analytics:

- Developed quantitative models supporting risk management for Bank of America's auto and home loan portfolios (the second-largest in the U.S.)
- Implemented big data analytics that successfully forecasted risk exposures during volatile market conditions
- Conducted meticulous model reviews that reduced operational and financial risks
- Ensured compliance with stringent regulatory standards

XL Catlin (2014-2015)

As Quantitative Risk Analyst:

- Developed and validated stochastic models for a \$40 million portfolio of equity, fixed income, and alternative investments
- Strengthened asset valuation methodologies for U.S.-based portfolios
- Enhanced precision in risk assessments and investment decisions

Genpact/Wells Fargo (2012-2014)

As Quantitative Analyst:

- Spearheaded advanced analytics and automation initiatives for Wells Fargo's Quantitative Corporate Finance team
- Supported corporate finance decisions in equity strategy and capital structure optimization
- Automated processes that reduced errors by 30-50% and addressed approximately 20 weekly issues

QcFinance India (2011-2012, 2015-2016)

As Big Data and Machine Learning Trainer:

- Spearheaded big data training and implementations for financial startups in Manhattan
- Integrated Hadoop and Spark ecosystems to establish real-time data lakes and machine learning pipelines
- Trained over 100 professionals in advanced data science techniques

7.1.3 Research and Publication Record

Mr. Joshi has established himself as a thought leader through a prolific and impactful publication record, demonstrating both breadth and depth in quantitative finance, AI, and risk modeling:

- Authored over 100+ pre-prints, thought articles, comments styled publications, generating a cumulative, snowballing impact that reflects sustained innovation and knowledge creation
- Received 0.5k-1k+ citations, highlighting the growing influence, relevance, and adoption of his work by U.S. researchers, government agencies, and policy institutions
- Many publications are indexed in Web of Research Commons, a supplementary database in Web of Science, which increases visibility but is not equivalent to core indexing and other scholarly databases, ensuring wide visibility, accessibility, and integration into ongoing research and practice
- Research emphasizes practical applications with direct relevance to U.S. financial markets, risk management, and emerging AI-based solutions, demonstrating the national importance of his contributions
- Through the snowballing effect of his publications, Mr. Joshi has contributed to the development of frameworks and methodologies that inform subsequent studies and policy recommendations, reinforcing his position as a knowledge generator and thought leader in his field

7.1.4 Professional Network and Collaborations

Mr. Joshi has built strong professional relationships with experts in his field:

- Recognition and praise from senior professionals at major financial institutions
- Collaborative relationships with academic researchers
- Engagement with the broader professional community through conferences and publications

USCIS Policy Manual Guidance. The USCIS Policy Manual (Vol. 6, Part F, Chapter 5) states that to satisfy the second prong of the Dhanasar framework, petitioners must demonstrate that “the person is well-positioned to advance the proposed endeavor.” This includes consideration of “the petitioner’s education, skills, knowledge, record of success

in related or similar efforts, and any progress towards achieving the proposed endeavor.” Evidence such as “letters from experts, awards, and previous accomplishments” can be used to show that the petitioner has the ability to advance the endeavor.¹

Web of Science Clarivate - Extensive Visibility of Work

Mr. Joshi has a total of 100+ publications indexed in the Research Commons, a supplementary database in Web of Science, which increases visibility but is not equivalent to core indexing.. Of these, 20 publications appear in the Web of Science Core Collection. His work has achieved widespread visibility and impact within the academic and professional community, reflecting a strong and consistent contribution to the fields of artificial intelligence and financial technology.

¹<https://www.uscis.gov/policy-manual/volume-6-part-f-chapter-5>

Chapter 8

Prong 3: Balancing Factors

8.1 National Benefit of Waiving Job Offer Requirement

Waiving the job offer and labor certification requirement for Mr. Joshi would provide significant benefits to the United States while protecting the interests of U.S. workers.

8.1.1 Unique Qualifications and Specialized Expertise

Mr. Joshi possesses a rare combination of skills and experience that cannot be easily replicated in the U.S. labor market:

- **Interdisciplinary Expertise:** Unique combination of advanced quantitative skills, AI expertise, and practical financial industry experience
- **Niche Specialization:** Expertise in applying generative AI to financial risk management, an emerging field with few qualified professionals
- **Proven Impact:** Demonstrated ability to deliver tangible results in systemically important financial institutions

The standard labor certification process is designed to protect U.S. workers from competition with foreign workers who possess similar qualifications. However, Mr. Joshi's unique combination of skills and experience means that there are few, if any, U.S. workers who could provide equivalent value to the national interest.

8.1.2 Urgent National Needs

The United States faces pressing challenges that require immediate attention:

- **Financial System Vulnerabilities:** Ongoing needs for enhanced risk management in systemically important institutions
- **Technological Transformation:** Rapid adoption of AI technologies in finance requiring specialized expertise
- **Workforce Skills Gaps:** Shortage of professionals with combined expertise in finance and advanced AI technologies

Delaying Mr. Joshi's contributions through the lengthy labor certification process would impede progress on these urgent national priorities.

8.1.3 Benefits Beyond a Single Employer

Mr. Joshi's proposed endeavor provides benefits that extend far beyond any single employer:

- **Public Research:** His scholarly publications advance knowledge and best practices that benefit the entire financial sector
- **Open-Source Tools:** Development of accessible tools and frameworks that can be used by regulators and smaller institutions
- **Workforce Development:** Training programs that enhance the skills of U.S. professionals beyond his immediate organization
- **Policy Contributions:** Research and analysis that informs regulatory approaches and policy decisions

These broad benefits would be delayed or potentially lost if Mr. Joshi were required to go through the standard labor certification process, which is designed for positions with specific employers rather than endeavors with national impact.

8.1.4 Impracticality of Labor Certification

The labor certification process is particularly impractical for Mr. Joshi's situation because:

- **No Appropriate SOC Code:** His interdisciplinary role doesn't fit neatly into existing occupational classifications
- **Unique Qualifications:** The specific combination of skills and experience required doesn't align with standard position descriptions
- **National Scope:** His proposed endeavor benefits multiple sectors and extends beyond any single employer's needs

USCIS Policy Manual Guidance. According to the USCIS Policy Manual (Vol. 6, Part F, Chapter 5), the third prong requires a showing that "on balance, it would be beneficial to the United States to waive the requirements of a job offer and labor certification." This involves considering "the national importance of the endeavor, the petitioner's qualifications, and whether the benefit to the U.S. outweighs the inherent national interest in protecting U.S. workers through the labor certification process."¹

Waiving the job offer requirement is in the **national interest** for the following reasons:

- **Public Benefit:** Mr. Joshi's open-source AI tools and open access can be used by the SEC to detect market manipulation, saving taxpayer resources.²

¹<https://www.uscis.gov/policy-manual/volume-6-part-f-chapter-5>

²SEC, *2024 Annual Report on AI in Enforcement* (Feb. 2024)

- **Urgency:** The DHS 2024 Strategic Plan prioritizes “AI for financial infrastructure security,” a field where Mr. Joshi is actively researching solutions.³ **His proposed collaboration and his research on adversarial AI threats is classified in the broader area as a “critical infrastructure priority” under CISA.**⁴
- **Impracticality of PERM:** His role involves **cutting-edge R&D** not captured by standard occupation codes. The DOL confirms “no prevailing wage data exists” for his niche GEN AI Risk Engineering.⁵ **A PERM process would delay his ongoing work with the U.S. Treasury’s Office of Financial Research.**⁶

8.1.5 National Interest Justification for PERM Waiver

Mr. Joshi’s contributions are **critical to U.S. economic stability, risk mitigation, and workforce re-skilling growth**, making the PERM process both impractical and contrary to the national interest:

- **Proposed Economic Stabilization Through AI Innovation:**
 - His risk modeling frameworks plan to propose directly addressed through open source publication the **U.S. Treasury’s financial stability AI monitoring programs**, enhancing systemic risk assessment capabilities.
- **Proposed Workforce Development at Scale:**
 - Created **industry-recognized training programs** that can upskill 1000+ U.S. professionals annually, helping US Citizens find AI roles and enhance their career. This will reduce outsourcing jobs to non-citizens abroad.
 - Mr Joshi also plans on partnering with the Department of Veterans Affairs to establish the **”Veterans in Financial AI”** initiative, creating direct pathways to high-value AI careers.
- **Urgent National Security Needs:**
 - Joshi’s work helps researchers address the NSA on **adversarial AI threat mitigation** addresses CISA-designated **”critical infrastructure priorities”**.⁷
 - Delaying this work via PERM would jeopardize **DHS’s AI security implementation timeline**.⁸
 - His role combines **cutting-edge R&D**, regulatory compliance, and workforce training - a combination **not captured by existing SOC codes**.

Conclusion: Mr. Joshi’s work plans to deliver **immediate, measurable benefits** to U.S. economic stability, workforce capacity, and financial system resilience. Requiring PERM would **unnecessarily delay** these national priorities while providing no protective benefit to U.S. workers, as his expertise is demonstrably unique.

³DHS, *AI Strategic Plan* (2024)

⁴CISA, *AI Threat Landscape* (2024)

⁵DOL, *Emerging AI Occupations Report* (2024)

⁶U.S. Treasury OFR, *AI Research Partnership Memo* (2024)

⁷CISA, *AI Threat Landscape* (2024)

⁸DHS, *AI Strategic Plan* (2024)

Unique Editorial Contributions Strengthening U.S. Economic Stability

Mr. Joshi's unparalleled expertise as a peer reviewer and editorial board member for **18+ international journals** and reviews work exclusively related to Risk pertinent to US Economy and market which directly enhances U.S. financial system resilience through rigorous knowledge validation. His editorial work focuses precisely on domains critical to national economic security:

Specialized Reviewing for Financial Risk Innovation

- **Journal of Risk and Financial Management (ISSN: 1911-8074)**: Evaluated manuscripts on Papers concerning Risk Models for US Banks.
- **FinTech (ISSN: 2674-1032)**: Certified peer reviewer for papers related to Credit Risk and Market Risk Models.

Peer Review Value Proposition

Mr. Joshi combines rare qualifications that make his editorial oversight indispensable. He has been achieved various peer review certifications.

: Peer review certifications from:

- Springer Nature (Fundamentals Modules I/II)
- Elsevier (Certified Peer Reviewer Course)
- Web of Science (Clarivate Training)

This unique intersection of **academic rigor, regulatory insight, and Wall Street implementation experience** enables Mr. Joshi to advance U.S. financial stability through peer review - a contribution that cannot be replicated through standard labor certification processes.

8.1.6 Open Ebook Publications Supporting Policy and Workforce Innovation

Mr Joshi is the author of two publicly available books on Barnes Noble: "*Agentic Gen AI For Financial Risk Management, ISBN: 2940179992974, Draft2Digital* " and "*Generative AI and Workforce Development in the Finance Sector, ISBN: 2940181548572, Draft2Digital*", which provide actionable insights at the intersection of artificial intelligence, regulation, and U.S. economic resilience. These books are designed not only for academics but also for decision-makers, educators, and practitioners across sectors.

Their availability on a mainstream platform like Barnes Noble ensures wide accessibility and underscores Mr. Joshi's commitment to public dissemination of research. These works translate advanced research findings into practical strategies, especially for policymakers navigating AI adoption, financial stability, and upskilling challenges in the national workforce.

The books contribute to the national interest in two major ways:

- **Policy Impact:** They provide a framework for applying generative AI in regulatory compliance, risk monitoring, and systemic oversight—aligned with federal priorities such as the NIST AI Risk Framework and DHS infrastructure protection goals.
- **Workforce Development:** The training-focused guidance in these texts supports the goals of the U.S. Department of Labor and CISA’s AI workforce initiatives by equipping professionals with accessible, structured pathways to integrate AI into their roles.

For details about the E-books, refer to Exhibit chap:exhibitebooks.

As such, these publications strengthen Mr. Joshi’s profile as a thought leader whose work is directly advancing both policy frameworks and labor competitiveness in the United States.

Mr Joshi’s peer-reviewed publications provide **actionable solutions** to challenges identified by U.S. policymakers, including:

1. AI-augmented financial stability mechanisms,
2. Scalable workforce training protocols,
3. Ethical guidelines for autonomous AI systems.

This body of work underscores Mr Joshi’s unique role in advancing U.S. leadership in AI innovation while safeguarding national economic and security interests.

8.1.7 Conclusion

Mr. Joshi’s unique expertise in financial AI and his standing as the top 10-15% researcher in this field provides immediate value to U.S. national interests. The PERM process is impractical given his niche specialization (no clear SOC code) and would delay critical work AI financial infrastructure deployments for different organizations. His open-source tools and publications already benefit U.S. researcher working in regulators and financial institutions without labor certification. Waiving the job requirement accelerates these contributions while protecting no comparable U.S. workers. His contributions strengthen U.S. financial infrastructure, align with federal priorities, and justify a waiver of the labor certification requirement. The national interest clearly favors waiver as his work is already being read and used by various professionals working on strengthening US Financial System.

Chapter 9

Documentation

This chapter provides a comprehensive overview of all exhibits submitted with this petition. Each exhibit serves to substantiate Mr. Satyadhar Joshi's qualifications, scholarly contributions, and the national importance of his proposed endeavor under the *Matter of Dhanasar* framework.

Professional Qualifications and Background

Exhibit ?? : Professional Resume demonstrates Mr. Joshi's extensive professional experience spanning over 10 years in quantitative analysis, AI applications in finance, and risk management at US financial institutions including Bank of America. This exhibit establishes his professional credentials and industry expertise.

Educational Credentials

Exhibit ?? provides official documentation of Mr. Joshi's advanced degrees and educational qualifications meeting EB-2 requirements.

Government and Federal Recognition

This section documents the extensive recognition and integration of Mr. Joshi's research by U.S. federal agencies and government-affiliated platforms. This multi-agency validation provides independent, third-party confirmation of the national importance and substantive merit of his work.

- **Exhibit ??: U.S. Department of Education – ERIC Database Indexing** – Five of Mr. Joshi's research papers are officially indexed in the Education Resources Information Center (ERIC), the primary digital library of the U.S. Department of Education. These papers address:
 - K-12 AI competitiveness (ED676035)
 - Rare earth elements curriculum (ED676036)
 - New Jersey AI leadership policy (ED678146)

– U.S. military AI reskilling (ED677111)

This inclusion demonstrates direct relevance to national education policy, STEM workforce development, and veteran transition programs.

- **Exhibit ??: U.S. Federal Reserve Board Citation** – Mr. Joshi’s research on generative AI in investment management is cited in the Federal Reserve’s Finance and Economics Discussion Series paper “Generative AI at the Crossroads: Light Bulb, Dynamo, or Microscope?” (Baily et al., 2025). This citation demonstrates that his work informs critical economic policy research at the highest levels of U.S. financial oversight.
- **Exhibit ??: U.S. Bureau of Labor Statistics (BLS) Utilization** – His research on AI and workforce disruptions is utilized in a BLS publication on workforce policy (DOI: 10.21916/mlr.2022.21). Backend metadata explicitly lists Mr. Joshi as a contributing author, confirming that his analysis directly informs federal labor market research and policy considerations.
- **Exhibit ??: U.S. Department of Energy – Science.gov Indexing** – Three of Mr. Joshi’s early peer-reviewed publications on secure multi-party computation and nano-RFID systems are indexed in Science.gov, the official portal for U.S. government science information managed by the DOE’s Office of Scientific and Technical Information. This indicates relevance to federally-funded research priorities across agencies such as NSF, NIH, NASA, and DOE.
- **Exhibit ??: Federal Agency Public Commentary (Regulations.gov)** – Mr. Joshi has submitted substantive expert commentary to seven federal agencies via the official U.S. rulemaking portal, Regulations.gov. These include the FDA, OSTP, CDC, ITA, FSIS, and NRC, demonstrating direct, on-the-record engagement in shaping federal policy across technology, health, trade, and safety domains.
- **Exhibit ??: U.S. RegInfo.gov Official Listing** – His comment on a Food Safety and Inspection Service (FSIS) docket is formally listed as one of only six public submissions on the official RegInfo.gov portal (managed by OMB/GSA). This scarce, selective inclusion underscores the substantive value of his expert input in the federal regulatory process.
- **Exhibit ??: U.S. Open Science Framework (OSF) Project** – Mr. Joshi’s comprehensive research on agentic AI in financial risk is hosted on the OSF, a platform operated by the Center for Open Science and supported by U.S. agencies including NSF and NIH. The project (DOI: 10.17605/OSF.IO/Z2EHR) is dedicated to the public domain (CC0 license), ensuring permanent, open access to critical AI risk management research for U.S. academic, government, and industry stakeholders.
- **Exhibit ??: U.S. Department of Commerce Library Visibility** – Multiple article records by Mr. Joshi are discoverable through the Department of Commerce Library’s broad search interface, indicating integration into federal library discovery infrastructure and accessibility to U.S. government researchers and policymakers.

Collective Significance: This multi-faceted government recognition—spanning education, labor, economic policy, science, health, and regulatory engagement—provides

compelling and independent validation that Mr. Joshi's work is of substantial merit and national importance. It demonstrates that his research is not confined to his corporate role but actively informs and is utilized by the U.S. government across critical domains, satisfying the *Dhanasar* framework's requirements for broad national impact.

Academic Repository Inclusion and Scholarly Integration

This section documents the formal inclusion and citation of Mr. Joshi's research within U.S. and international academic repositories, demonstrating its scholarly merit, accessibility, and practical utility across the educational and research ecosystem.

- **Exhibit ??: Harrisburg University Digital Commons** – Contains multiple publications by Mr. Joshi archived in this institutional repository, including works on AI leadership and agentic AI governance. This inclusion validates the scholarly merit and accessibility of his research to academic communities, with some publications receiving early citations.

text

- **Exhibit ??: Michigan Technological University Digital Commons** – Includes Mr. Joshi's collaborative engineering research on plasmonic solar cells, permanently archived and accessible to students, faculty, and researchers in renewable energy and engineering disciplines.
- **Exhibit ??: Smithsonian/NASA ADS Database (Harvard)** – Contains Mr. Joshi's foundational work on secure multi-party computation for defense applications, indexed in this astrophysics and interdisciplinary research database, reflecting high academic caliber and cross-disciplinary relevance.
- **Exhibit ??: Stanford University Libraries (SearchWorks)** – Catalogs his conference paper on secure statistical computation from ICSTE 2009, ensuring long-term preservation and discoverability through one of the world's known academic library systems.
- **Exhibit ??: Massachusetts Institute of Technology (MIT) Center for Transportation & Logistics Citation** – Shows that Mr. Joshi's optimization research has been referenced in publications from MIT, indicating recognition and utilization at a premier U.S. research institution.
- **Exhibit ??: U.S. National Institutes of Health (PubMed Central) Citation** – Provides independent, third-party validation that Mr. Joshi's published work has been cited by other researchers in a peer-reviewed article indexed in the official NIH repository, demonstrating relevance within federally recognized scientific literature.

Collective Significance: This extensive academic integration—spanning institutional repositories, premier university libraries, federal research databases, and curriculum citations—demonstrates that Mr. Joshi's work is recognized, preserved, and utilized by the U.S. educational and research ecosystem. It provides tangible evidence that his contributions extend beyond any single employer or professional role, fulfilling

Scholarly Publications and Research Impact

Exhibit ??: **Authored Books** includes two open-access books on Generative AI applications in workforce development and financial services, showcasing Mr. Joshi's ability to synthesize complex technical concepts for practical industry applications.

Exhibit ??: **Selected Thought Leadership Papers** contains Mr. Joshi's research papers focused on enhancing U.S. financial market stability through AI applications. These publications demonstrate his contributions to the field and establish his expertise in financial risk management.

Indexing and Scholarly Validation

Exhibit ??: **Econ Papers Indexing** shows inclusion in economics research databases, indicating recognition within the economics research community.

Exhibit ??: **Research Commons, a supplementary database in Web of Science, which increases visibility but is not equivalent to core indexing** demonstrates global research visibility through inclusion in a premier academic discovery platform used by researchers worldwide.

Exhibit ??: **SSRN Repository** shows early dissemination of research through a well used social science research network, indicating engagement with academic communities prior to formal publication. Author rank of 4,648 out of 2,583,020 places Mr Joshi in the top 0.18% of researchers worldwide (99.82nd percentile), representing an elite, globally recognized standing. This metric, which positions me among the top 5,000 authors in a multi-million researcher database, serves as a quantitative validation of high-impact influence in the field.

This is further supported by a robust preprint dissemination record, with my work garnering over 23k downloads and 35k views on MDPI preprint platforms (indexed in WOS), demonstrating significant reach and engagement within the scientific community prior to formal publication.

Exhibit ??: **ResearchGate Profile** provides evidence of professional networking and research sharing within global academic communities, with demonstrated readership and citation metrics.

Exhibit ??: **Semantic Scholar Indexing** shows inclusion in AI-focused academic search engines developed by research institutions, validating relevance to AI research communities.

Exhibit ??: Industry citation and referencing.

Conclusion

Collectively, these exhibits provide comprehensive evidence supporting all three prongs of the *Dhanasar* framework. They demonstrate Mr. Joshi's substantial scholarly contributions, the national importance of his proposed endeavor, his unique qualifications to advance this work, and the clear national benefit of waiving labor certification requirements. Each exhibit has been carefully selected to address specific USCIS criteria and

provide verifiable, objective evidence of Mr. Joshi's eligibility for the National Interest Waiver.

Chapter 10

Exhibits